Mechanical Ethics · Casebook

Case studies and technical analysis

Institutional information disclosure

Wildfire protective action

Technical employment design

Database recovery

Accessible reading services

Five-case reading edition · 27 October 2026

Four source-based studies and one constructed design study.
With comparative analysis, source notes and editorial history.

A supplement to Mechanical Ethics 1.0.
These studies do not establish field effectiveness.

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Markdown source pinned to 9acb06fb9ca07814076b7462cca0ee2f08ae4cbb. Study texts are unchanged. The original three-case comparison and dated editorial history are retained alongside the later five-case comparison; historical statements describe their own edition. Legal and other source cutoffs remain those stated in each study. This document changes reading format, not evidence or rights. External sources require a connection.

Case 1. IRS–ICE information disclosure: requirements, implementation and residual exposure

Taxpayer information and immigration enforcement in the United States, April 2025 to September 2026

> **What this case is.** A worked example for readers of Mechanical Ethics v1.0. The history comes from public court opinions, statute text, an Inspector General's report and a Congressional Research Service summary, all listed at the end with what each one supports. It uses no interviews and no private material. It is not legal advice and takes no side on immigration policy. The court decisions described are **preliminary**: none of the sources used reports a ruling on the final merits, and the latest of them is dated 17 September 2026. Where the case proposes something rather than reports it, the text says so.

Problem and analytical scope

The IRS sent 47,289 address records to ICE on 7 August 2025 under a statutory disclosure exception. The relevant appellate decisions subsequently distinguished an agreement likely lawful on its face from a procedure likely unlawful as used. The engineering problem is the gap between substantive disclosure conditions and executable checks: the presence of fields and a successful identity match did not establish every condition governing release.

Taxpayer confidentiality also affects the system that produces the records. The appeals court quotes an earlier ruling describing privacy as "fundamental to a tax system that relies upon self-reporting". The case examines the documented disclosure process separately from the proposed feedback through trust, filing behaviour and data quality. The latter is not measured at population level in this record.

Analytical questions

The analysis distinguishes four questions: what information was requested; what conditions governed disclosure; what the implemented procedure established; and what restrictions or remedies followed the transfer. Request volume, matching success, legal sufficiency and downstream use are separate variables. The record supports some of these more strongly than others.

System boundary and evidence access

Boundary chosen for this case: the route by which information moved from the IRS to ICE under one statutory exception, §6103(i)(2), between April 2025 and September 2026. It also covers that route's documented effect on people's willingness to engage with the tax system.

Left outside on purpose:

- the merits of immigration law and of any removal order;
- what ICE did with any address;
- other data-sharing arrangements (the Massachusetts case also names the Social Security Administration);
- anyone's private motives.

Entities followed:

- the roughly 46,965 people whose addresses were sent;
- the roughly 1.28 million people listed;
- taxpayers who file without a Social Security number;
- IRS staff who process disclosures;
- the IRS and ICE as institutions, and Treasury and Homeland Security leadership;
- TIGTA, the courts and Congress;
- the organisations that sued and the people they serve.

Apertures and accounts. In the sources used here, each party sees part of the route, and none of them shows the whole of it.

Party	Can see	Cannot see
IRS	its own records; ICE's request file	whether any listed person is really under criminal investigation
ICE	its own removal records and investigations	IRS records, except what is sent
TIGTA (Inspector General)	IRS processes and data	Did not independently validate the DHS-supplied data or assess legality; this describes its review scope, not an inability to access any such data
Courts	the administrative record, sworn declarations, argument	anything outside the record; at this stage, only likelihoods
Organisations that sued	changes in their clients' behaviour	whose addresses were sent
Affected people	their own fear or decision	in general, whether their own address was sent (I found no notice to them in these sources)

Party	Can see	Cannot see
This case's author	public documents only	sealed filings, the implementing agreement, TIGTA's redacted findings, the interview transcript some tellings rely on

Chronology

All dates below come from the sources listed at the end. Where two sources disagree, the disagreement is noted.

Date	Event	Source
1976	After Watergate, Congress restricts the IRS's sharing of return information with other agencies (26 U.S.C. §6103)	D.C. Circuit opinion, Sept 2026, citing the 1976 Senate report
26 Nov 2024	An IRS rule published in the Federal Register states that no provision of the U.S. Code authorises disclosing returns or return information "for enforcement of immigration laws"	Quoted in the D.C. Circuit opinion, Sept 2026
20 Jan 2025	Executive Order 14161 directs steps to identify and remove people present unlawfully	TIGTA report; agreement text
17 Feb and 7 Mar 2025	Two lawsuits filed in Washington, D.C. (Center for Taxpayer Rights v. IRS; Centro de Trabajadores Unidos v. Bessent)	TIGTA report, App. II
7 Apr 2025	Treasury (for the IRS) and Homeland Security (for ICE) sign a Memorandum of Understanding. The IRS's Chief Privacy Officer declares that no request has yet been made under it	Agreement as filed in court; IRS declaration
Before 17 Apr 2025	The IRS manual describes individual review of each such request by a disclosure manager	D.C. Circuit opinion, Sept 2026
18 Apr 2025	Implementing agreement signed (not public in the sources used here)	TIGTA report, App. III
12 May 2025	District court declines to block the agreement (Centro)	TIGTA report
Early June 2025	ICE asks for information on "the full alien population" of more than 7 million people	District court finding, quoted by the D.C. Circuit
27 Jun 2025	ICE's Acting Director asks the IRS for the last known address of about 1.28 million people	D.C. Circuit opinion; TIGTA ("over 1.2 million")

Date	Event	Source
1 Jul 2025	IRS staff are given the "green light" to process the request	D.C. Circuit opinion
Jul to Aug 2025	The IRS builds an automated "Data-Exchange Procedure" to match ICE's file to its records	D.C. Circuit opinion; TIGTA
7 Aug 2025	The IRS sends 47,289 records (addresses as of 6 August)	TIGTA; IRS declaration via CRS and the court
Oct 2025	TIGTA sends an email alert: ICE's open safeguard findings need follow-up	TIGTA report
21 Nov 2025	District court stays the procedure and bars further disclosure except in strict compliance with §6103(i)(2), with advance notice to the court	D.C. Circuit opinion; CRS; TIGTA
Nov to Dec 2025	TIGTA raises matching concerns; the IRS reviews manually and confirms them	TIGTA report
23 Jan 2026	The IRS notifies Homeland Security and asks for help to remediate	TIGTA report
5 Feb 2026	District court in Massachusetts bars ICE and DHS from using the disclosed information	CRS
11 Feb 2026	The IRS's Chief Risk and Control Officer files a declaration admitting some disclosures answered deficient requests	CRS; TIGTA; D.C. Circuit opinion
24 Feb 2026	D.C. Circuit (Centro): the agreement itself likely complies with the statute	Centro opinion; CRS
26 Feb 2026	District court: "A zip code is not an address"	Quoted in the D.C. Circuit opinion
4 Jun 2026	Treasury Inspector General for Tax Administration (TIGTA) publishes its evaluation	TIGTA report 2026-IE-R010
8 Sep 2026	D.C. Circuit (Center for Taxpayer Rights): the procedure as used likely violates §6103(i)(2); the district court's order is affirmed	Opinion and judgment, No. 26-5006
7 Oct 2026	First Circuit argument scheduled in the Massachusetts appeal; outcome not checked for this case	CRS (as of 17 Sep 2026)

A small discrepancy, recorded rather than resolved. The September 2026 opinion twice cites the district court's "Order of Nov. 21, 2025", but its closing paragraph affirms the "order of November 1, 2025", and the judgment repeats that date. The Centro opinion, TIGTA and CRS all give 21 November. This case uses 21 November and treats "November 1" as a slip in the opinion's text.

Units and reported counts

The reported counts have different denominators and units. They must not be used interchangeably.

Count	What it measures	Source
more than 7 million	people in ICE's early-June request about "the full alien population"	district court, quoted by the D.C. Circuit
about 1.28 million	people in ICE's 27 June request for last known addresses	ICE letter, as described by the court; IRS declaration via CRS
47,289	records the IRS sent to ICE	IRS declaration; court; TIGTA
about 46,965	individuals those records correspond to	TIGTA's own analysis
42,695 (90.3%)	disclosed records matched by taxpayer number, without matching ICE's supplied address against the IRS address	IRS declaration via CRS; court ("more than 90%")
"less than 5%" of 47,289	people for whom ICE's address was potentially incomplete or insufficient (for example "Unknown Address", or no street)	IRS's own estimate

Three cautions follow.

- **Requested is not disclosed.** About 1.28 million people were asked about. About 47,000 had an address sent. A figure of seven million also circulates. The record contains an earlier ICE request about more than 7 million people, and in the sources used here I found no record that the IRS processed it.
- **What was sent:** the last known address only, with no date attached. The IRS told TIGTA that no other tax information was shared.
- **One witness behind most numbers.** The 47,289 figure appears in the court opinion, the CRS summary and the TIGTA report, but each traces back to the IRS's own records and declaration. In the sources used here, the only separate count is TIGTA's own analysis of IRS data (46,965 individuals). As the book warns, "three files repeating one witness are not three independent accounts" (Ch. 9, line 302).

Disclosure mechanism

What carried the information, step by step:

1. **The promise:** §6103(a) makes returns and return information confidential unless Title 26 expressly allows disclosure. TIGTA puts the public side plainly: taxpayers "have the right to expect" that information is not disclosed unless authorised by them or by law.
2. **The legal channel:** §6103(i)(2). On a written request from an agency head that gives four things, the Treasury "shall disclose" return information other than

taxpayer return information. The four are the taxpayer's name and address, the taxable period, the statutory authority, and the specific reason the information may be relevant. Disclosure may go only to officers and employees "personally and directly engaged" in the criminal investigation or proceeding, "solely" for that use. Under §6103(i)(2)(C) a taxpayer's identity, which includes their mailing address (§6103(b)(6)), is not treated as taxpayer return information.

3. **The agreement:** it tracks the statute, sets safeguard duties for ICE (IRS Publication 1075, training, annual reports, breach notice) and states that it "does not confer any rights or benefits on any third party". So the taxpayer is not a party to it.
4. **The request:** a data file of about 1.28 million rows and one cover letter. The district court found that the file named the same point of contact for every request. The letter gave one reason for all of them: the information "may contain address information which is potentially at issue" in investigating failure to depart (8 U.S.C. §1253(a)(1)).
5. **The procedure:** preprocessing tested field presence and a weak address-format proxy. Matching then followed different paths depending on whether ICE supplied a taxpayer number. On the taxpayer-number path, the IRS address could be returned without matching ICE's supplied address against it. That describes the matching path; it does not establish a legal requirement that the two addresses be identical. Passing a format screen, obtaining an identity match and establishing the substantive disclosure conditions were different claims. The appeals court found that the procedure did not require ICE to submit an address and, on the taxpayer-number path, did not confirm that the supplied information plausibly identified the taxpayer's name and address.
6. **The output:** a last known address, undated. The appeals court reasoned that without a date the address cannot be compared with the date of a removal order. So it cannot show that someone stayed after the order, which is the very offence cited.

Couplings, from documented to proposed:

- Documented: filing a return puts a current address in IRS records, and the procedure made that address reachable by an enforcement agency.
- Documented at the stage of a preliminary finding: news of the policy reduced engagement with at least one taxpayer clinic. The Center for Taxpayer Rights' director swore to a "drastic" fall in cases from taxpayers without Social Security numbers, and to moving "nearly 10 percent" of the clinic's operating expenses into outreach. The courts accepted this as likely harm. They reasoned from the "predictable effect of Government action on the decisions of third parties", and compared it with a citizenship question on the census depressing participation.
- Proposed, not measured: lower willingness to file could reduce the completeness of tax data, and with it public revenue and the quality of decisions that rely on that data. I found no systematic filing statistics for 2026 that test this. The book's

caution applies: a questionnaire or monitoring system "can also change behaviour being measured; records are not passive windows" (Ch. 12, line 476).

Requirements and implemented checks

The analysis distinguishes three claims (ME Chapter 9): "a check specified, a check performed and a check capable of finding this failure" (Ch. 9, line 300).

Statutory condition or earlier administrative practice	What the procedure checked	Could it find the failure?
name **and address** of the taxpayer	address field contains any five or nine digits; on a taxpayer-number match, the supplied address was not matched against IRS records	No. Entries in the format "NA NA, AZ, 999990000" passed (TIGTA's illustration)
recipient "personally and directly engaged"	point-of-contact field "not empty"	No. One person was named for every request
the "specific reason" disclosure is relevant	one cover letter for 1.28 million requests	No. The court called the reason "more circular than specific"
individual handling under the earlier IRS manual, rather than a separate statutory element	until April 2025 a disclosure manager reviewed each request and could offer less intrusive options	Replaced by "a mass, automated review of millions of records at the press of a button"

Agreement and implementation. The agreement specified the statute's checks; Centro found it likely lawful on its face. The procedure performed checks, but they could not detect a request that failed the statute's conditions. In the court's account of the argument, the government acknowledged "concededly unlawful" transfers. Its counsel called the missing address check "an error in implementing the policy". The agreement's facial compliance did not establish that the requests or implementing procedure complied. Performed checks did not establish the required substantive conditions.

This is also the book's machine-speed point (Ch. 12, lines 478 to 482). The statute allows disclosure only to officers personally and directly engaged in each investigation. The procedure turned that condition into a field that only had to be non-empty. "A human name in an approval field does not establish that the person saw the relevant evidence."

TIGTA also found potential matches lost through differences in name or address formatting. That does not show those requests were legally eligible for disclosure. It does mean that correcting the procedure is a two-sided task: reject deficient requests without silently treating harmless formatting differences as a different person. A matched identity is still not permission to disclose. The report does not quantify all such exceptions, so no overall error rate is inferred here.

Process model and diagnostic implications

The following abstraction separates the operations described in the source record. It is an analytical model, not a reconstruction of source code or an additional legal test.

Let r denote one row in the submitted request file, considered with the cover letter and other request-level material that applies to it. It does not denote a unique person or an output record. Multiple rows may concern one person; the available counts do not establish a one-to-one mapping between rows, people and released records.

Let F(r) denote passing preprocessing; M_b(r) an identity match on branch b (taxpayer-number or non-taxpayer-number); L(r) satisfaction of the applicable disclosure conditions for the proposed release; and O(r) the event that at least one record is released in response to r. The intended condition is $O(r) \Rightarrow L(r)$. L is a substantive classification requiring authorised legal interpretation and facts, not a Boolean calculated by this chapter. F and M_b are processing results. Their conjunction does not establish L without evidence that the tests cover its requirements.

Stage or branch	Evidence reported in this record	Unestablished inference
Shared preprocessing	Field presence and address-format proxy	That the supplied address and stated reason meet the substantive conditions
Taxpayer-number matching	Identity match could lead to output without comparing the supplied address with the IRS address	That identity resolution establishes every disclosure condition; address equality is not asserted as a statutory test
Non-taxpayer-number matching	A distinct matching path; TIGTA identifies formatting-related possible missed matches	That this path's complete executable coverage has been reconstructed, or that a missed match was a legally eligible request
Release and recipient	Output records and common contact information are reported	That a non-empty contact field establishes personal and direct engagement, or that each proposed recipient/use is authorised

The table maps documented gates to their evidential limits. It does not claim access to the implementation's source code or a complete branch specification.

The taxpayer-number branch illustrates why coverage must be assessed by branch, not by the presence of a check somewhere in a workflow. A test performed on one path does not constrain another path that reaches the same output. Conversely, a field can be examined during preprocessing without its substantive adequacy being established. The relevant diagnostic question is therefore which conditions each reachable output path actually enforces, not whether the system contains an address-related operation.

A complete row-level assessment would distinguish four outcomes: a row supporting a compliant request with or without a release, and a row supporting a deficient request with or without a release. Rows whose substantive classification cannot be resolved must remain a separate unresolved category. Person-level and output-record rates would require their own mappings and denominators. The published disclosure counts do not populate this classification. The IRS estimate of potentially deficient addresses is narrower than all possible grounds of deficiency. TIGTA's formatting exceptions identify possible missed matches, not legally eligible requests. Neither can be used as the overall false-acceptance or false-rejection rate.

This distinction also prevents a misleading repair target. Maximising matches could increase deficient releases; maximising rejections could withhold requests that the statute requires the agency to answer. A useful validation set must separate identity resolution from legal eligibility and include both kinds of error. Where the legal classification is contested, the uncertainty belongs in the case label rather than being concealed as an engineering ground truth.

The disclosed-record count is an output measure, not a complete exposure measure. The approximately 46,965-person count removes some record duplication, but does not measure later access, onward disclosure or use. Likewise, a court restriction creates an enforceable constraint; it is not an observation of every subsequent access. Evaluating containment would require appropriately authorised records of custody and use. This case does not have those records.

Control and exposure

Actor	Could change	Exposed to
Treasury and Homeland Security leadership	whether the agreement existed, and its terms	political and legal accountability
Treasury and IRS lawyers	whether processing began (they gave the go-ahead)	the court's findings
The IRS units that built and ran the procedure	the procedure's checks	staff exposure under §7213 (felony for willful unauthorised disclosure)
ICE leadership	the content of each request	the court's view that the requests were deficient
Courts	whether the procedure could keep operating	none of the direct harms
Congress	the statute itself (CRS notes Congress could amend it)	none of the direct harms
Affected people	mainly their own choices: whether to file, seek help or sue	location disclosed; fear; loss of help with time-limited tax matters

The asymmetry is the book's point about the clock and the door (Ch. 6, line 156). The people who could change the route were not the people exposed to it. Individual IRS employees appear on both sides of that line. Their use of the procedure was "not optional", according to the court, and yet the criminal penalty for willful unauthorised disclosure falls on individuals.

Timing of disclosure and restriction

The book asks us to name each loss and give it its own clock (Ch. 6, lines 170 to 178). The protective effect has to arrive strictly before the loss hardens: T(protect) < T(harden).

1. **Disclosure of the first batch.** It hardened on 7 August 2025, when the addresses left the IRS. Copies can be returned or destroyed, as the agreement provides, but a disclosure cannot be undone in a comparable form. The first court order restricting the procedure came on 21 November 2025. **For this batch, prevention failed.** The order restricted further disclosure; it could not prevent the completed August transfer.
2. **Use of those addresses.** This is a separate loss with a separate clock. The Massachusetts order of 5 February 2026 barred ICE and DHS from using the information. The IRS also told TIGTA that DHS and ICE had confirmed they would not act on the data while the lawsuits were pending. TIGTA reports this as the IRS's statement, and I found no independent check of it in these sources. **Whether any address was used between 7 August 2025 and 5 February 2026 is UNKNOWN in the sources used here.**
3. **Willingness to engage.** Trust in a promise is built by keeping it over time (Ch. 6, line 162). The court records a fall in engagement at one clinic after the policy was reported, and it stresses that taxpayers' own deadlines have "no do-overs". How fast trust recovers is UNKNOWN.
4. **Safeguards while data is held.** A 2023 IRS safeguard review of ICE left findings open; the number is redacted. According to the IRS, ICE filed no corrective plan for three half-year periods (January 2024, July 2024, January 2025). It filed one in July 2025 without implementation dates, and by May 2026 had not filed the January 2026 update. The IRS's on-site review set for April 2026 was postponed by a partial government shutdown. This risk lasts as long as ICE holds the data.
5. **The legal clock.** In the source record ending on 17 September 2026, the rulings described are preliminary and a **hold** is in place. A hold creates "a second exposed party" (Ch. 6, line 196). Here that is the government's interest in criminal investigations, which the IRS says the order "impedes and delays". That cost belongs in the account even though the court found it outweighed.

Evidence independence and contestability

- **TIGTA**, the Inspector General: independent of IRS management and working under professional standards. It triggered the IRS's self-review: its concerns in November and December 2025 led the IRS to re-examine its own disclosures. Its limits are its own: it relied on IRS data, did not validate ICE's data, did not assess legality, made no recommendations, and redacted its safeguard findings.
- **The IRS's own declaration:** an admission against the agency's interest, so strong evidence that errors occurred. It is also the source of nearly every count. The book warns that two records can share one source (Ch. 9, line 298).

- **The courts:** adversarial, working from a record and sworn evidence, but so far ruling only on likelihoods. The two D.C. Circuit decisions look opposite but address different things. Centro reviewed the agreement on its face. Center for Taxpayer Rights reviewed the procedure as it was used.
- **The Congressional Research Service:** a careful secondary summary for legislators, not an investigation.
- **Congress:** letters on 12 February 2026 and a hearing on 4 March 2026, at which the IRS's chief executive declined to comment, citing litigation.
- **The affected people:** mostly absent from the record. Section 7431 provides a damages route subject to liability conditions and exceptions; compensation is not automatic. Using that route also depends on discovering the disclosure. The readable agreement pages describe a disciplinary-action notice route. Separately, §7431(e) requires notice for specified criminal charges or proposed disciplinary determinations; whether those triggers applied here is not established. The route that reached a court was organisations suing on their own account, and the Center's injury was precisely that its clients had stopped coming forward. The book anticipates both points: the affected person "may choose not to expose themselves" (Ch. 9, line 328), and gathering many partial accounts is collective evidence (line 332). A department "cannot certify its own answerability" (line 316). Here an outside witness and a court did the work the procedure could not.

Competing interpretations

The strongest reading for the government:

- The statute says the Secretary "shall disclose" on a compliant request, and addresses are not taxpayer return information. The appeals court accepted that reading of the agreement in February 2026.
- Failure to depart after a final removal order is a crime, and enforcing it is a lawful duty.
- The errors were implementation errors. By the IRS's estimate fewer than 5% of the people disclosed had potentially deficient ICE addresses, and only addresses were shared.
- The evidence of a chilling effect comes from declarations by organisations that are parties to the case, not from measured filing data.
- Treating "blockers" as heroes ignores that refusing a lawful request is also a breach of duty.

The strongest reading for the plaintiffs:

- The taxpayer-number path, used for about 90% of disclosed records, did not require an ICE-to-IRS address match. Weak format screening did not establish the substantive sufficiency of the supplied address or the other required conditions. This does not make address equality an independent statutory requirement.

- The district court found it "facially implausible" that one contact could be "personally and directly engaged" in about 47,000 criminal matters, let alone 1.28 million.
- The stated reason was the same circular sentence for everyone.
- The IRS's own earlier position was that no provision allowed disclosure for immigration enforcement. Until 2025 it read §6103(i)(2) to prohibit disclosing an address when nothing else was requested.
- The district court went further and said the record "raises an inference" that the criminal-investigation basis was pretext. That is a preliminary inference by one court, not a finding on the merits. The appeals court recounted it but rested its holding on the statutory defects. This case does not adopt it.

What the cited courts accepted in the source record through September 2026: the agreement is likely lawful on its face, and the procedure is likely unlawful as used. Both rulings are preliminary.

Alternative procedures

These are declared alternatives for the same request, not a ranking. No numerical harm difference is calculated: the sources do not supply an evidenced common scale and counterfactual for these alternatives. Which course the law requires is for courts and Congress.

Alternative	What it protects	What it costs
A. Individual review, as in the IRS manual before April 2025	each request is checked by a person who can contact the prosecutor and offer less intrusive options	slow at scale; whether it could handle 1.28 million requests lawfully is UNKNOWN
B. The procedure as built	speed	the performed proxy checks and matching paths did not establish the required substantive conditions (Requirements and implemented checks)
C. Proposed: automation that validates every statutory field	speed with real checks	feasibility UNKNOWN; scale may still conflict with "personally and directly engaged", which the courts read narrowly
D. The court-order route, §6103(i)(1)	a judge decides first (an ex parte order is required for returns and taxpayer return information)	slower; it is a different channel for different information
E. Refuse all such requests	maximum confidentiality	on Centro's reading of "shall disclose", a breach of the duty to answer compliant requests

Alternative E matters for the "blocker" question. On Centro's reading, a gate that refused everything would break the law in the other direction. In the book's terms, the protective control the statute asks for is a check able to tell a compliant request from a deficient one.

Correction ownership and acceptance

Protective corrections reported within the source cutoff:

- The district court stayed the procedure and required notice before any further disclosure. Owner: the court. Effect: the IRS told TIGTA it had shared no further data; the first batch was not recalled.
- The Massachusetts court barred ICE and DHS from using the data. Owner: that court. It was under appeal at the source cutoff.
- TIGTA's alert, then the IRS's manual review, notice to DHS and sworn admission. Owner: TIGTA, then the IRS.

The correction-theatre test (Ch. 9, line 336) asks which rule, interface or authority actually changed, and whether someone outside the failing route can check it. An outside authority did change: the court's notice requirement. Whether the script itself was fixed is UNKNOWN. At argument in May 2026 the government's own counsel said, "I'm not aware of what action IRS has taken" to fix it. Notifying DHS and asking for help is evidence of receipt, not of protective effect.

Constructive corrections. These are proposals, offered for the named owners to weigh. Mechanical Ethics does not decide them.

- **Assess and address the open safeguard findings.** Proposed owner: the IRS Office of Safeguards with the appropriate DHS counterparts and legal advisers. Identify what protective action, conditions or restrictions the findings warrant under the applicable duties and court requirements, and what evidence would establish that action's completion. TIGTA's October 2025 alert called for follow-up; this proposal does not turn that alert into an established blanket rule or a finding that safeguards have since been repaired. Under §6103(p)(4), safeguards are a condition of receiving return information; the Secretary may refuse disclosure until satisfied that the requirements have been or will be met. Whether to exercise that power here belongs to the authorised decision-maker under applicable law and court requirements.
- **Specify and demonstrate the corrected disclosure route.** Proposed owners: the IRS Office of Privacy, Governmental Liaison and Disclosure with Chief Counsel and the responsible engineering team, subject to the applicable court requirements. Legal sufficiency, a working check and permission to disclose are separate decisions. A code repair would not by itself authorise a transfer. The following would make the proposed repair inspectable; none is reported completed here.

Proposed work	Evidence a competent reviewer would need	What would still not follow
State which request conditions must be established and who may decide disputed cases	An authorised, source-linked explanation distinguishing missing information, uncertain matches and unresolved substantive questions	That filling every field establishes a lawful request

Proposed work	Evidence a competent reviewer would need	What would still not follow
Map those conditions to preprocessing, both matching branches and the final output	A branch-by-branch account showing that a successful match cannot silently bypass a required condition	That a taxpayer-number match supplies every other necessary justification
Examine both inappropriate release and unnecessary rejection	Controlled examples: a deficient request stops before disclosure; a harmless formatting variation is examined without silently changing the person or treating a possible match as an authorised release	That fewer rejections prove better matching, or that more rejections prove better protection
Assign acceptance and follow-up within legitimate authority	A named responsible role, recorded unresolved defects, a review point and evidence of what was actually changed, available to the appropriately authorised reviewer	That an announcement, an internal pass or this proposed table proves completion

The scrutiny itself must respect taxpayer confidentiality: use non-identifying test material where appropriate and keep necessary real records within lawful custody. The proposal is not a request to publish personal data, grant new access or resume transfers. Each failure to meet an acceptance condition returns a specific unresolved question to its legitimate owner rather than producing an automatic legal verdict.

- **Establish applicable notice duties and any further notification options.** Owner: Treasury and the IRS with legal advice. Section 7431(e) contains specified mandatory triggers; further notice requires its own legal assessment. Whether a trigger applied or notice occurred here is UNKNOWN.
- **Reduce the main UNKNOWN with aggregate data.** Counts of returns filed with Individual Taxpayer Identification Numbers by year would show whether such filing fell, though not why. Owners: IRS Statistics of Income or the National Taxpayer Advocate. Only aggregates should be used: per Ch. 12, line 476, measurement must not become a new surveillance route.
- **Settle the statute.** Owner: Congress, which could tighten or loosen §6103(i)(2). Both are legitimate options; this case does not choose between them.

Residual exposure and prospective capability

Restrictions on future disclosure do not restore the prior custody of information (ME Ch. 10, line 348). Residual effects and unresolved exposures include:

- about 47,000 addresses held outside the IRS, subject to a court restriction on use while an appeal was pending at the source cutoff;
- whatever fear and lost engagement occurred, including tax relief that some people may have missed because they stayed away;

- one clinic's diverted budget;
- the IRS staff's legal exposure;
- the open question of whether any address was used.

New prospects opened by the same events:

- a public, judicial reading of what "personally and directly engaged" means at scale;
- TIGTA's continuing audit of safeguards across all of the IRS's data-sharing agreements;
- a documented record that later readers and institutions can check.

Reporting errors

Claim	Qualification supported by the record
"Handed over the records of more than a million people"	About 1.28 million people were requested; 47,289 records (about 46,965 people) were disclosed, address only and undated.
"Overrode confidentiality"	The statute contains a narrow channel for disclosure in criminal investigations, and on its face the agreement likely fits it. The requests and implementing procedure still had to satisfy the substantive conditions; performed checks did not establish that they did.
"Officials who resisted were protecting the law"	TIGTA reports that officials had legal concerns, citing news media. Correction involved the courts and the Inspector General, with the agency reviewing and admitting errors after the Inspector General raised concerns. The November 2025 court restriction preceded the February 2026 admission. A blanket refusal of compliant requests would conflict with the cited reading of the statute.
"The courts ruled it illegal"	Two preliminary rulings, about two different things, pointing in what look like opposite directions. The cited decisions do not settle final merits; the First Circuit appeal was pending at the source cutoff.

The record distinguishes the existence of a legal disclosure channel from satisfaction of its conditions in individual requests.

Relation to Mechanical Ethics

Question the book already asks	Where
Was the check specified, performed, and able to find this failure?	Ch. 9, line 300
Are these independent witnesses, or one source repeated?	Ch. 9, lines 298 and 302
Is access the same as authority to disclose?	Ch. 9, line 306

Question the book already asks	Where
Can an institution certify its own answerability?	Ch. 9, line 316
Can the affected person safely come forward?	Ch. 9, line 328
Do many partial accounts reveal one mechanism?	Ch. 9, line 332
Did the rule, interface or authority change, or only the announcement?	Ch. 9, line 336
Who controls the clock, and who is exposed while it runs?	Ch. 6, line 156
Did protection arrive before the loss hardened, for each loss separately?	Ch. 6, lines 170 to 178
What does a hold cost, and whom?	Ch. 6, line 196
Is a correction a restoration?	Ch. 10, line 348
Does measuring change the behaviour measured?	Ch. 12, line 476
Is a name in a field meaningful oversight at machine speed?	Ch. 12, lines 478 to 482

What this case adds, and what it does not. The book already holds each of these questions. Its scenes are mostly constructed for explanation (Notes, line 594), and its attributed historical cases are brief. This case follows one real institutional chain end to end with sources, as far as the evidence goes. It does **not** establish the arrow from a broken promise to less truthful filing to worse public data. That arrow remains a proposed mechanism: in the sources used here it rests on sworn declarations about one clinic and on a legal premise the courts repeat. Nor does the vocabulary decide anything. "A description alone cannot grant rights, produce moral priorities, prove motives or make a remedy happen" (Notes, line 580).

Unresolved questions and domain authority

- **The final legal merits**, including the First Circuit appeal after this case's source cutoff: courts own this. Legal analysis is a stronger method than this case.
- **Whether any disclosed address was used before 5 February 2026**: ICE, the courts and the DHS Inspector General, to whom TIGTA said it would send its concerns.
- **Whether affected people were or will be notified**: Treasury and the IRS.
- **Any population-level effect on filing or revenue**: tax economics and IRS statistics. I found no systematic 2026 data.
- **The content of the implementing agreement, TIGTA's redacted findings and sealed filings**: not public in the sources used here.
- **Whether the pretext inference holds**: the courts, on the merits.
- **The immigration-enforcement cost of the hold**: ICE and DHS; it was not measured in these sources.

Sources

Retrieved 26 October 2026 unless stated. File hashes and the claim-by-claim check are in the companion source receipt.

1. Center for Taxpayer Rights v. IRS, No. 26-5006 (D.C. Cir. 8 Sept 2026). Opinion (Pillard, J., for Millett, Pillard and Wilkins, JJ.) and judgment. https://storage.courtlistener.com/recap/gov.uscourts.cadc.42769/gov.uscourts.cadc.42769.2191763.0.pdf and https://storage.courtlistener.com/recap/gov.uscourts.cadc.42769/gov.uscourts.cadc.42769.2191760.0.pdf
2. Centro de Trabajadores Unidos v. Bessent, No. 25-5181 (D.C. Cir. 24 Feb 2026), opinion by Senior Circuit Judge Edwards (Srinivasan, C.J., Millett, J., Edwards, S.J.), 167 F.4th 1218. https://media.cadc.uscourts.gov/opinions/docs/2026/02/25-5181-2160705.pdf
3. Treasury Inspector General for Tax Administration, The IRS Provided Addresses for Nearly 47,000 Persons to Immigration and Customs Enforcement, Report 2026-IE-R010, 4 June 2026. https://www.tigta.gov/sites/default/files/reports/2026-06/2026ier010fr.pdf
4. Congressional Research Service, Updates on Litigation over the IRS-ICE Information-Sharing Agreement, Legal Sidebar LSB11413, updated 17 Sept 2026, read via the EveryCRSReport mirror. https://www.everycrsreport.com/files/2026-09-17_LSB11413_111e02a83756aa234badbaf17530964afcc9c3fc.html (official page: https://www.congress.gov/crs-product/LSB11413)
5. Memorandum of Understanding between Treasury/IRS and DHS/ICE, with the IRS Chief Privacy Officer's declaration, filed 7 Apr 2025 as Exhibit A, ECF 30-1, Centro, No. 1:25-cv-00677 (D.D.C.). Copy hosted at https://files.epi.org/uploads/IRS_ICE_MOU-April-2025.pdf (several pages are images with no extractable text).
6. 26 U.S.C. §6103 (especially (a), (b)(6), (i)(1), (i)(2)), §7213 and §7431, via the Legal Information Institute. https://www.law.cornell.edu/uscode/text/26/6103, https://www.law.cornell.edu/uscode/text/26/7213, https://www.law.cornell.edu/uscode/text/26/7431
7. Mechanical Ethics v1.0.0, tagged source, line numbers as cited. https://github.com/markgoodbody-bit/mechanical-ethics/blob/v1.0.0/working-editions/v1.0/ME10_DEV4R_CORR2_Source_Preserved_Master.md?plain=1

Source receipt · Return to the casebook

Case 2. Camp Fire evacuation and temporary refuge

Case B for the Mechanical Ethics v1.0 supplement. Research and interpretation as of 26 October 2026; separate from the frozen released book. Historical findings, interpretation and proposed planning work are distinguished below. This is a worked inquiry, not instructions for choosing a refuge during a live fire.

An evacuation plan can show a road that exists, a notification can be issued, and people can start travelling. None of those events establishes that they have reached safety. The difficult question is what happens when fire changes the available journey faster than people and institutions can complete it.

The Camp Fire makes that question concrete. NIST's 2023 NETTRA publication records the evacuation of about 40,000 people from the affected California communities, 85 fatalities, and the use of 31 temporary refuge areas involving more than 1,200 civilians. Its reconstruction draws together notifications, traffic, fire conditions and rescue activity rather than treating evacuation as one decision. These are historical findings from a particular event, not numbers produced by Mechanical Ethics. [1]

The purpose here is to follow one relationship: **a warning must connect to an action that becomes protective while that action remains feasible**. When one route closes, the remaining work does not necessarily end. It changes. The question may shift from leaving the community to reducing immediate exposure, and then to making onward movement possible.

Event and system boundary

The fire began on 8 November 2018. NIST's fire-progression study documents rapid spread, strong winds and long-range spotting affecting Concow, Paradise and Magalia. A single moving front is therefore an inadequate picture of the changing danger: fire could establish ahead of a traveller's position. [2]

The NETTRA study identifies congestion, blocked roads and direct fire exposure during evacuation. Refuge areas were used within that disrupted movement. They were not evidence that ordinary evacuation had remained available to everyone. [1]

This case's chosen boundary runs from noticing danger through protective movement, temporary refuge and onward evacuation. It also asks how the resulting knowledge can affect future preparation. The wider causes of ignition, individual legal liability, land-use decisions, insurance, ecological damage and long-term recovery require their own investigations. Leaving them outside this particular comparison does not make them unimportant or resolve anyone's responsibility.

The affected scope includes people who received messages, people who did not, people unable to move without assistance, responders exposed while helping them, and people elsewhere competing for finite transport or response capacity. A person absent from the

available observations has not thereby been shown to have had a usable exit. This is an analytical boundary choice, open to challenge; it is not a new estimate of any group's historical exposure.

Observations and source dependence

A resident can observe nearby smoke, traffic and the message actually received. A dispatcher may see notification requests and communications. A responder may know that a particular junction has become dangerous. Incident management needs an account broad enough to coordinate resources, but breadth does not guarantee that its information is current at every location. These are roles in the analysis, not invented testimony from particular people.

Radio messages, location records, maps and observations can carry useful information between these positions. A physical road carries people; a message carries an account of whether that road can be used. Improving one carrier does not automatically repair the other. An accurate closure report is valuable even though it does not clear the obstruction. A cleared road is valuable even though someone who has not received usable directions may not reach it.

The 2026 paper's Methods describes interviews with responders and officials, supplemented by logs, tracking and media; it did not survey residents or directly study their decision-making. That makes it a substantial reconstruction with an identifiable aperture, not every affected person's complete account. [5]

To challenge a specific claim, ask which record supports it. Does an order log show a request, actual dissemination, receipt by the relevant person, or subsequent action? Were two timestamps independently anchored or reconciled from a shared account? A second publication reusing the same reconstruction is not a second independent event observation. The NIST reports, their later research synthesis and their news summary must not be counted as three replications of the outcome.

Timing and outcome definitions

The relevant clock is not simply time since somebody pressed an alert button. NIST's timing account includes assessment, transfer and dissemination, followed by people's preparation and movement. Its stages are an aid to investigating elapsed time, not a reason to assume every stage occurs neatly in sequence. [3]

Consider four endpoints in this case's interpretation:

Question	Effect that must be established	What an administrative timestamp cannot establish
Did the warning help?	Relevant information became usable by someone still able to act	That issuing an order informed that person
Was outward evacuation feasible?	The person could traverse the relevant route before hazardous conditions defeated it	That a road appeared on a plan

Question	Effect that must be established	What an administrative timestamp cannot establish
Did refuge provide protection?	Exposure was reduced for the people concerned for the necessary period	That an area had been named or occupied
Was the next stage completed?	People reached an appropriate onward destination as conditions permitted	That entry into temporary refuge ended the incident

For a specified person and route, the comparison needs a shared starting event and evidenced bounds: time until a protective effect versus time until that course no longer protects against the specified threat. ME1 Chapter 6 preserves this necessary strict-before relationship. It does not supply the missing fire model, travel-time distribution or exposure threshold. A single deadline for an entire town would conceal differences in location, mobility, information and route conditions. [ME]

Hardening is equally specific. A blocked road can close one journey without proving every alternative closed. Fatal injury cannot be undone by a later report. Avoiding immediate injury may leave later needs unresolved. Future planning can still be improved after the historical opportunity to protect a particular person has passed. Calling an investigation late must therefore name the loss it cannot now prevent, rather than dismiss its possible value for future people.

Protective-route model and limits of identification

A protective journey can be represented as a sequence of conditional transitions: warning reception, interpretation, preparation, mobilisation, movement and arrival at a location that remains tenable. This is an analytical decomposition, not a fitted evacuation model. Failures can occur at several transitions and share causes. Loss of power, for example, may affect both communications and other activities; the factors should not be treated as independent probabilities.

For a specified person or group, t_a is receipt of a usable warning, τ_p the elapsed time from receipt to departure, and τ_m the duration of the realised journey. Then $t_{arrive} = t_a + \tau_p + \tau_m$ is an accounting identity for that journey, not a prediction. It assumes these events can be identified; departure before receipt or without a warning requires a different starting event.

Feasibility must be assessed along the trajectory. Divide the journey into segments i, with entry time e_i and exit time x_i. For each segment, the group's traversal interval $[e_i, x_i]$ must lie within the interval or intervals during which that segment is accessible and tenable for that group. If a segment has one known closing threshold u_i and remains usable beforehand, $x_i < u_i$ is the corresponding conservative clearance condition. Waiting at junctions or in queues also occupies locations and must be included. A segment that closes after the group has passed does not make its later arrival infeasible.

Thus t_arrive < min_i(u_i) is not a generally necessary condition. It can be imposed as a conservative special-case screen, but would reject some journeys that clear each segment in time. Nor does it test a destination's continued tenability. At a temporary destination, conditions must support the group throughout occupancy, from entry until onward departure or rescue, with capacity and assistance available. Endpoints alone do not establish acceptable accumulated smoke or heat exposure; specialist criteria are required.

The inequalities identify a modelling requirement, not operational thresholds. No segment timings, exposure limits or safe margins are estimated here. Where closure times or travel durations are uncertain, a nominal positive margin is not a demonstrated robust route. Assessments would need bounded uncertainty or a justified scenario analysis, including correlated failures.

Earlier receipt can increase preparation time, but its effect on the realised trajectory depends on behaviour, assistance and network demand. Simultaneous earlier departures can alter queues; unavailable transport can prevent departure altogether. Shifting only the historical warning timestamp while holding all subsequent movement fixed would therefore require justification.

Refuge changes the route and the exposure profile; it does not remove the timing problem. An assessment must include entry conditions, tenability during occupancy and an onward movement or rescue path. A site could reduce immediate flame exposure while retaining smoke exposure or dependence on responders. Those are different outcomes and may have different responsible operators.

The retrospective categories reported by NIST do not estimate the prospective effect of publishing a refuge map. That intervention can change departure, destination choice and expectations before an incident. Its effect would require evidence about those behavioural responses, site assessment, maintenance and incident conditions. The Marin appendix establishes published planning products and some programme ownership; it does not supply these outcome measurements.

An engineering assessment would therefore need linked, time-resolved evidence: warning receipt rather than activation alone; departure and assistance requests; route states and traffic; hazard conditions; refuge occupancy and duration; and subsequent movement and outcomes. Incomplete or inconsistent timestamps would need explicit uncertainty bounds. These are requirements for further specialist analysis, not data claimed to have been collected for this supplement.

The main work's clock distinction is useful here because a single event can close one route while leaving another temporarily available. Determining those windows requires fire, transport and human-factors methods. The vocabulary alone cannot establish a safe site, departure threshold or operating plan.

Outcome definitions and inconsistent summary claims

A tempting first account is: **the warnings came too late, so earlier warnings would have solved the evacuation**. That compresses several different failures into one. NIST's ESCAPE explanation separates the activation of a notification process from preparation and transport. It expressly warns that the start of dissemination is not the time at which the population has been informed. [3]

The corrected account is narrower. Earlier effective notice may help where a usable movement window exists. It cannot by itself establish transport, passable routes, assistance or sufficient remaining time. A notification improvement and a viable evacuation are different achievements. The analyst must therefore examine the full journey and contingency arrangements, rather than infer the remedy from the alert timestamp.

There is a second correction about outcomes. NIST's April 2025 news account describes improvised refuge decisions as saving more than 1,200 people. The 2026 researchers instead attribute prevention of death for **at least 261 civilians** and reduced danger for **at least 961 others**. These are retrospective classifications, not an experimental estimate. Refuge involvement is not converted into a separately established life saved. [4, 5]

That correction changes the comparison. Refuge involvement, reduction of exposure, survival and a causal estimate of deaths prevented cannot be substituted for one another. Nor can totals from refuge and rescue categories simply be added: the supplied sources do not establish that they represent disjoint people.

NIST publishes both accounts and is the appropriate owner to reconcile or annotate the earlier summary. The narrower research categories are retained here; the difference is not independently resolved.

Control, exposure and coupled resources

The person in the vehicle does not control the entire road network, available buses or incident response. Equally, an incident commander does not control wind, every ignition, or every individual's ability to move. A useful account locates an actor's actual capacity instead of assigning omnipotence to the most visible institution.

There are also coupled costs. Holding traffic away from a dangerous route can expose people waiting elsewhere. Assigning a crew to one location can make it unavailable at another. An earlier movement instruction may alter congestion. These are questions a comparison must investigate, not historical conclusions established here. Neither a count of options nor the word precaution allocates the resource or settles the competing duty.

This separates causal explanation from blame. A vehicle can contribute to a queue without its driver having authority to redesign evacuation arrangements. A researcher can identify a defect without possessing operational command. Assessing culpability requires the applicable duties, knowledge and conduct; ME's descriptive categories do not confer them. [ME]

Temporary refuge: function and operating limits

The 2026 paper distinguishes immediate refuge from traffic-risk management and describes active supervision and later movement between or out of refuge locations. It also warns about inadequate separation from fire, dependence on responders and smoke exposure. Its proposed pre-identified Temporary Fire Refuge Areas are not guaranteed safe places; the paper reports no TFRA design standard at the time of publication. [5]

NIST's 2025 ESCAPE update develops preplanning for no-notice events and designated fire-refuge contingencies. It explicitly treats these alternatives as a fallback, not a substitute for evacuation where evacuation remains feasible. [4, 6]

The constructive implication is a maintained protective arrangement, not merely more signs. For a proposed site, a competent local planning process would have to establish suitability for relevant conditions, practical access, usable information, capacity and a subsequent course when conditions change. This is a proposed line of inquiry for responsible specialists, not a finding that an existing parking lot satisfies it.

Maintenance matters because a previous assessment can cease to describe the present place. Ownership, vegetation, access, surrounding development or available resources can change. A sign may outlast the conditions that justified it. The proposal therefore includes an accountable review and correction route, not an assumption that installing a label creates a durable capability.

Competing planning hypotheses

Pre-designation has two competing risk implications. Both require site-specific assessment.

Name suitable refuge areas before the next fire. On this reading, the Camp Fire shows both what improvisation under fire costs and what it achieved. Responders quickly directed people to open spaces with lower fire intensity. The 2026 paper reports that this ad hoc use of 31 refuge areas saved at least 261 civilian lives and reduced the risk to at least 961 others, a retrospective classification as noted above. Its stated implications are that communities should include temporary fire refuge areas in their fire response plans, and NIST's 2025 update emphasises pre-designating them. Waiting for a complete standard would leave the next no-notice fire to improvisation again. [4, 5]

Do not let a named place stand in for a way out. On this reading, the same sources set limits that a public label can hide. As noted above, NIST treats refuge as a fallback for when evacuation can no longer be completed, not a substitute for it, and the 2026 paper reports no design standard for such areas, along with risks from inadequate separation from fire, smoke and dependence on responders. A site announced in advance might be trusted beyond what anyone has assessed, or treated as a destination by people who could still have left. [5, 6]

Both readings accept the same historical record. They differ on which risk a community should carry now: improvising again, or endorsing a place that has not been shown to work for the people expected to use it. This case does not settle that for any real

community, and neither does Mechanical Ethics. It is a decision for the local emergency-management and fire authorities in the table below, with the people who would have to use the site.

The following evidence would distinguish the planning hypotheses:

- an assessment of a specific site against the conditions it would face;
- evidence on whether naming a site in advance changes when people leave, which the sources used here do not measure;
- an owner with the resources to keep the site usable.

These dependencies are unresolved in the available source set.

Alternative interventions and outcome measures

The useful comparison is not between perfect knowledge and the actual disaster. It is between defined courses available to specified actors under plausible conditions.

Course to examine	Possible gain to investigate	Cost or failure the comparison must retain
More effective earlier notification	More usable preparation or movement time	Receipt may fail; movement capacity may remain inadequate
Prepared assistance and transport	A route for people unable to use ordinary departure options	Staffing, dispatch delay and competition for vehicles
Professionally assessed refuge contingencies	Reduced exposure when an outward route cannot be completed	Site failure, smoke, access, capacity, responder burden and onward movement
Rely on improvisation alone	Flexibility during an unanticipated event	Suitable places and available help cannot be assumed

These are prospective comparison questions, not four tested treatment groups. The historical reconstruction supports investigating them; it does not estimate each course's effect in a new community. A numerical **ΔH remains UNKNOWN**: there is no justified common scale here combining fatalities, injuries, smoke exposure, responder risk, disruption and expenditure against an evidenced counterfactual. Reporting separate outcomes is more informative than inventing a total. [ME]

In particular, do not take 85 deaths and subtract a refuge headcount to create a number of preventable deaths. Those numbers describe different things. An adequate quantitative assessment would require specialist scenario modelling, explicit assumptions and uncertainty, plus a legitimate decision about the values being compared.

Implementation responsibilities

The following is a **proposed allocation to be confirmed locally**, not a claim that these offices already accepted a task or that this chapter establishes their legal duties.

Work	Legitimate owner to establish	Evidence of an achieved effect
Test movement and refuge options	Local emergency-management and fire authorities with relevant engineering, transport and accessibility expertise	A scenario-specific assessment, limitations and accountable decision
Make an adopted arrangement usable	Responsible local agencies and site custodians	Checked access, maintained conditions, understandable information and assigned resources
Coordinate during an incident	Authorised incident management and responders	Location-specific information reaching someone able to adapt the response
Challenge omissions before an incident	Responsible planning authority, with affected residents and independent specialists	An accessible response explaining what changed, what did not, and why
Preserve and improve the knowledge	Research organisations, professional bodies and operational owners in their distinct roles	Findings incorporated, tested or rejected with reasons, rather than merely downloaded

NIST supplies research and guidance in these sources. Publication alone does not constitute a local evacuation order, site certification, budget allocation or implementation. Emergency managers' specialist methods are the stronger starting point; the case does not propose a parallel ME command structure.

A consequential review should let someone unable to use the assumed route reveal that problem without first becoming an evacuation modeller. It should also let responders challenge a plan that quietly allocates them to incompatible tasks. Accessibility, residents' practical constraints and safe handling of personal accounts belong in the design of that review; collecting identifiable disaster testimony is not necessary for this written case.

Residual effects and future capability

Immediate survival is a different outcome from restoring a home, health, livelihood or community. Those domains remain outside the outcome estimates in this case. ME1's residue distinction keeps the question open without manufacturing a personal loss ledger or implying that recovery is impossible. [ME]

The positive prospect is the capability to prepare, maintain and adapt protective arrangements together. Research can carry knowledge beyond the responders who first improvised. A future community may become better able to recognise a missing route, obtain specialist help and correct an unworkable plan before an emergency. That is a prospective capability, not an observed benefit attributable to this supplement.

The practical handover question is small enough to be answerable: **for the people least able to use the assumed evacuation route, which responsible owner can show the workable alternative and how it will remain workable?** If existing ESCAPE work already answers it, use that work. If no competent owner or adequate evidence can answer it, record the actual gap rather than certify the plan through vocabulary.

Existing specialist methods and limits of contribution

NIST's specialist work already integrates fire behaviour, notification, movement and refuge. It does not need ME to discover that safety depends on more than sending a message. Adding a second conceptual worksheet could waste scarce planning time, imply competence the author does not possess or obscure technical uncertainty with reassuring labels.

That objection is substantial. ME earns no efficacy claim from this case. The modest possible contribution is helping a nonspecialist follow distinctions across a handover: an account versus a usable route; notification versus protective effect; control versus exposure; recorded survival versus a causal counterfactual; research publication versus maintained implementation. Those distinctions are already present in ME1 Chapters 5, 6, 8, 9, 10 and 12, not new primitives introduced by this fire. [ME]

An actual comparative use test could still find **EXISTING_METHOD_BETTER** or **NO_USE**. For example, if an authorised planner's ordinary ESCAPE process already exposes the relevant constraints more clearly, this restatement adds no practical value. No planner or resident has applied this case in the present work, and no such test result is claimed.

The earned result is a source-bounded worked account with corrected claims. Whether this account improves a real decision remains an external question.

Sources and exact reading boundaries

1. **Historical reconstruction:** Maranghides and colleagues, A Case Study of the Camp Fire: Notification, Evacuation, Traffic, and Temporary Refuge Areas (NETTRA), NIST TN 2252, July 2023. Official publication and abstract. The abstract and NIST life-safety project account were checked. The complete 278-page technical report and supplement were not fully read in this task.
2. **Fire progression:** Maranghides and colleagues, A Case Study of the Camp Fire: Fire Progression Timeline, NIST TN 2135, February 2021. Official abstract. Scope: spread and investigation summary; no independent reconstruction of its event database here.
3. **Specialist timing method:** NIST ESCAPE, Understanding the evacuation timeline, sections on notification activation, dissemination and movement. This is the operational comparator, not a method derived from ME.
4. **Public summary and correction target:** NIST, Updates Critical Wildfire Evacuation and Sheltering Guidance, 24 April 2025. Its broad lives-saved wording is not silently transferred into the more specific research classification.
5. **Later research synthesis:** Eric Link and Alexander Maranghides, Lessons learned from community-scale use of temporary refuge areas during the Camp Fire, International Journal of Wildland Fire 35(8), WF25312, 20 August 2026. Publisher HTML Abstract; Methods; Results: Management of TRAs; Discussion: Performance and limitations, and Pre-planned temporary fire refuge areas were inspected. The

NIST publication record corroborates metadata and abstract, not an independent replication.

6. **From research to preparation:** NIST ESCAPE programme, sections introducing TFRAs and no-notice events, and official course entrance. No complete PDF, training completion or local site assessment claimed.

[ME] Released conceptual source: Mechanical Ethics v1.0.0, exact original manuscript. Git blob `2caa284150f14630f0ed0fca5d41a3530925dec5`, SHA-256 `1315a2c2b500ae2308afd7070d5b3919b23d1390870e65eea54bbc057e6e37a8`. Relevant anchors: lines 124/132 (route); 156/170/174/178/194 (clocks and competing resources); 250/260/276 (comparison); 288/298/302 (witness and custody); 354/366 (residue); 462/464/480 (distributed responsibility and interruption); 586/622 (specialist limits). These passages were reread; no fresh complete-manuscript reread is claimed.

Public sources were retrieved on 26 October 2026. Large report PDFs could not be fully inspected through the available retrieval paths; no PDF-page or figure verification is claimed. This account therefore supplies section-level HTML locators, not invented page citations. No original figure, survivor quotation, private case data or third-party report reproduction is included.

Study 3. Supervision capacity in entry-level technical employment

Positive capability formation · England/UK policy, checked 26 October 2026.

This case combines a documented policy question with an explicitly **constructed** small-employer situation. No firm, entrant or supervisor took part. No job was offered, funding eligibility decided, system changed or outcome measured. The purpose is to make a possible first job concrete enough for the people who would carry it to challenge it.

Policy context

The UK Government's 21 October Digital and Technologies Jobs Plan sets an ambition of approximately 200,000 entry-level jobs over ten years. Its existing approach includes job design, local coordination and meaningful work with supervision. An ambition is not a delivered job. [C1]

The June DSIT/LinkedIn snapshot reports weak entry-level hiring alongside a weak wider market, with variation between occupations. It explicitly does not establish AI as the cause. Its platform data do not represent every route into work. A convincing story about disappearing junior tasks cannot replace that causal uncertainty. [C2]

Nor is this an unoccupied policy field. The Early Careers Jobs Alliance already addresses task analysis, organisational design, progression and employer–union pilots. These are existing institutional routes to investigate, not discoveries made by this casebook. [C3] The question left for our constructed firm is narrower: **can it support one real paid beginning without making somebody else silently finance it?**

System definition and assumed task

Suppose a small IT-services firm is considering a paid beginner role. It has a qualified supervisor and a non-production test environment, but limited spare supervision. The firm cannot authorise a newcomer to change customers' live systems. A manager thinks AI-assisted diagnosis has reduced some routine work that beginners once performed. That last statement is a hypothesis inside this example, not a finding about an actual employer or the economy.

A candidate task is to reproduce a fault in a synthetic service, distinguish observed behaviour from a suggested explanation, and produce a test that another person can run. The beginner may use AI to propose a cause, but must be able to say what was checked and what remains unknown. The supervisor owns approval of any later live change. This is not automatically a suitable apprenticeship, productive job or employment arrangement: the appropriate employer, provider and legal processes would have to decide those matters.

The proposed outcomes include paid useful work, transferable diagnostic skill, appropriate escalation and progression to a different task. Attendance and a correct AI-assisted answer do not establish those outcomes. Working relationships and the entrant's experience also require evidence rather than inference from task completion.

System boundary and resource incidence

If the boundary contains only the beginner's output, the project looks cheap: one workstation, one task, one completed test. Expand it just enough to include the people and resources that make that output possible.

Affected party	What they may gain	What the first account can hide
Adult entrant	Pay, skill, confidence, a credible next step	Travel, access needs, insecurity, work that teaches little
Supervisor	A capable colleague and better tests	Preparation, interruptions, review, recovery and displaced duties
Existing staff	Shared knowledge and a future colleague	Work shifted onto them or an expectation of unpaid help
Employer	Useful work and a stronger skills pipeline	Cash commitment, uncertain demand and continuing responsibility
Customers and other users	Better diagnosis and reliability	Data exposure, poor changes or reduced service attention
Training or placement partner	A viable route into work	Unsupported promises and duplicate assessment work
People not selected	A future opportunity or fair competition	Selection barriers, inaccessible recruitment and exclusion from the record

This is an authored boundary. It does not assign identical rights or decision powers to everyone in the table. A client can control access to its systems; that does not make it the judge of a worker's learning. A beginner's account is necessary evidence about their experience; it does not give them the firm's entire staffing budget. The point is to keep the different decisions visible.

Necessary operational transitions

A route to a useful first task would require the following connections to work:

1. A manager can commit actual pay and protected supervisory capacity.
2. An appropriate recruiter or partner reaches a willing adult who can use the offer.
3. The supervisor supplies a safe, comprehensible task with legitimate access.
4. The entrant can practise, ask and disagree without being punished for reasonable uncertainty.

5. The result reaches a competent reviewer, who distinguishes a lucky answer from a learned method.
6. The next task and continuing support give the capability somewhere to go.

Each join has a carrier: a funded role, an accessible offer, a usable test environment, time with another person, a work sample and an agreed next step. An approved plan cannot substitute for a cancelled hour of help. A completed learning record does not show that the entrant can perform a different task. An AI transcript is an account of a tool interaction, not an independent witness that the fault was understood.

Some controls preserve the possibility rather than obstruct it. Keeping customer credentials outside the exercise can allow a beginner to learn safely; removing the access boundary to make the role look more “real” may destroy the conditions under which it was responsible to offer it. Equally, a sandbox filled with busywork can be safe and still fail to supply useful learning or paid productive purpose.

Capacity and scheduling constraints

This design study can be made more specific without inventing employer data. Define a planning interval and let H be the supervisor's available paid hours, E existing committed work, P task preparation, I instruction, R review, and U unplanned assistance or rework. A necessary capacity condition is $E + P + I + R + U \leq H$. All terms must use the same interval and avoid counting the same activity twice. No values are supplied because no employer was observed.

An allocation that satisfies the condition only after assuming $U = 0$ is sensitive to interruptions and novice error. An allocation based on average U can still fail during concentrated demand. A second condition concerns scheduling. Partition the planning horizon into slots narrow enough to represent the task's support needs. For each slot k, let A_k be competent supervisory capacity remaining after existing duties, and D_k the instruction, review and assistance that must occur in that slot. A proposed schedule requires $D_k \leq A_k$ in every slot, without allocating the same person's time simultaneously to incompatible work. Hours supplied by an unqualified substitute do not count as competent capacity.

Some tasks need uninterrupted attention or a response within a specified interval; a sum of fragmented availability cannot establish either. Those constraints require a feasible assignment of people to tasks, including preparation and handover time. For uncertain demand, the design must specify the disruptions it is intended to tolerate, a qualified fallback and the response when that capacity is exceeded. No probability of overload or acceptable response time is estimated here. A nominal hour on Friday does not resolve a blocked or unsafe task on Tuesday. The relevant design includes a fallback owner and a bounded task that can pause without transferring risk to customers or pressuring the entrant to proceed unsupported.

Training funding and employer supervision costs

Training funding is one input to feasibility. It does not establish the availability of productive supervision or the affordability of the employment arrangement.

For English apprenticeships, the 2026–27 rules require appropriate employer support and supervision (66.1). They distinguish eligible, standard-linked training/mentoring under specified conditions (107.6) from wages and generic line-management support that cannot be charged as apprenticeship training costs (112.1, 112.11). Training plans, reviews and complaints routes already exist (100–103). The register condition in 107.6 includes stated exceptions. These rules concern this funding route, not every ordinary starter job. [C4]

A training allocation is not evidence that all the employer's supervisory work is paid for. Before offering the role, the budget owner must identify which hours the ordinary route already covers, which hours remain an employer cost, and which existing commitments will move. If the honest answer is that the supervisor will absorb the difference at night, this version of the role is not yet supported. Renaming those hours “mentoring” does not supply capacity or settle eligibility.

Claude Code's internal AI premise challenge brought this distinction into the project before this case was written. It also challenged the cost of adding another assessment beside existing reviews. This case therefore proposes using an existing work/training review where suitable, rather than requiring a new ME form. That is an editorial correction from source and criticism, not a successful field intervention. [C5]

Funding and capacity are separate constraints. Let C_w be wages and employment costs, C_s supervision costs, C_t eligible training costs, C_o other operating costs, and F the funding actually confirmed for the arrangement. A cash assessment must allocate F only to permitted cost categories and identify who covers the remainder. An unconfirmed allocation is not a reduction in required cash. The source rules distinguish eligible training from wages and generic management; this expression does not decide eligibility for any individual activity.

Opportunity cost must also be retained. A supervisor can remain within paid hours while displacing existing work. That can reduce service capacity or move work to another employee. The financial ledger and the hours ledger therefore answer different questions. Both are needed before interpreting a role as supported rather than merely affordable on a narrow training budget.

Timing constraints

The firm has a budget decision. The entrant has a possible start date and other commitments. The supervisor has current deadlines and limits on attention. Practice has its own rhythm: a long gap can interrupt what the person is learning. The organisation's future skills pipeline changes over a much longer period.

Those clocks cannot responsibly be compressed into one deadline. Moving the start date might protect supervision but cost the entrant another opportunity. Starting immediately might preserve pay while producing poor support. A firm may need to decline this version now, or work through an established partner, rather than manufacture urgency as an excuse for a weak offer.

For a specified avoidable loss, protection has to become effective before that loss hardens. But no durations have been measured here, and missing a proposed start date is not evidence of permanently lost employability. The case supplies neither a numerical timing margin nor a universal instruction to pause hiring.

Alternative designs and cost allocation

Alternative	A plausible gain in this constructed setting	What must still be shown
Recruit an experienced person	Meet an immediate service need	Whether this is affordable and whether entry routes can be supported elsewhere
Fund the bounded paid beginner role	Create useful work and a learning route	Pay, actual supervisory hours, meaningful task and continuing support
Use an established apprenticeship or partner route	Draw on specialist training and existing review arrangements	Suitability, eligibility, employer commitments and uncovered costs
Delay or decline this offer	Avoid a promise the firm cannot maintain	An honest explanation and whether another usable route can be offered

These are candidate comparisons, not recommendations to a real business. The proper baseline is what this employer could actually have chosen, not an imaginary world with unlimited mentors. The affected scope includes the staff whose existing work is displaced. The horizon needs to reach beyond the first successful demonstration.

No scalar **ΔH** is calculated. We have no observed pay, hours, learning outcomes, client risk or defensible conversion between them. A future comparison could record those quantities separately; it should not turn an entrant's dignity or a mentor's fatigue into invented units. A supported positive prospect is not merely the negative of a quantified harm score.

Alternative interpretation

An experienced provider could say: “We already do this; your additional language takes time from supervision.” They might be right. The case is also biased towards a technical practitioner willing to consider a role: it cannot speak for an excluded entrant, an exhausted supervisor or a worker who distrusts the review process. A safe synthetic task may not supply the commercial demand needed to sustain employment. AI may be incidental to the firm's actual barrier.

The honest response is not a longer checklist. Find the existing method and ask what, if anything, this account changes. If it creates no better action, reveals no omitted burden and offers no clearer supported prospect, record **NO_USE** for this application. If it exposes one real unresolved join, pursue that join with its owner. Neither outcome is a verdict on every future use of the language.

Observation and verification

In a real voluntary test, first ask the practitioner how they normally decide whether such a role is feasible. Record that answer before introducing our questions. If their existing approach already exposes the relevant constraints, credit it. A clearer conversation is useful, but not automatically evidence that ME changed the decision. [C5]

Possible evidence then has different jobs. A budget record can support a pay commitment; a rota can support allocated time; an appropriately bounded work sample can support what was attempted. A later unfamiliar task might test transfer of learning. The entrant can contest whether the work was usable and fair. These sources can all be partial.

Five documents copied from one manager's judgement are not five independent witnesses. Nor should an unhappy account be discounted because it conflicts with the institution's dashboard. Independence includes who could observe the failure, who controls the record and what it costs to disagree. Necessary evidence should not become a pretext for collecting a worker's private life.

Output quality is not a sufficient learning measure. A correct diagnosis may have been supplied by AI or repaired by a supervisor. Evidence of increasing capability would require performance on a different but relevant task, an account of the assistance used, and observation over time. A single successful ticket cannot separate independent capability from external support. Conversely, requesting help is not itself failure; appropriate escalation is part of the specified task.

The proposed arrangement has at least three independent acceptance questions: can the employer sustain its obligations; can the entrant use the arrangement and refuse inappropriate work; and does the work develop relevant capability? Failure of one is not repaired by success in another. A useful pilot would record planned and actual supervisory demand, displaced work, completion and rework, assistance, and the entrant's account. It would also specify who can reduce scope or stop the arrangement while preserving employment obligations. No such pilot has been conducted here.

Correction authority and implementation

In this proposed arrangement the employer's authorised budget owner would own pay and resourcing. A competent supervisor would own task boundaries and review. A relevant provider or broker would own its actual training and escalation responsibilities. The entrant would retain ordinary rights and a legitimate voice in the experience, not an

obligation to prove the experiment worthwhile. Applicable employment, equality, funding and safety requirements must be resolved through competent domain processes; this chapter does not determine them.

The funding boundary this case relies on is not the firm's to change. The 2026–27 rules record a machinery-of-government transfer of responsibility for apprenticeships from the Department for Education to the Department for Work and Pensions, and they point to Skills England for each standard's minimum off-the-job training. [C4] If firms like this one repeatedly find that the supervision a beginner needs falls outside what the rules can fund, that pattern belongs with those owners, through the consultation routes they run. It is best carried by the people who would supervise and pay for the hours, not only by bodies speaking for them. In automatic pension enrolment, the Institute for Government found that consulting trade associations was not a sufficient way to get input from those who would have to administer the system, and that regulations had to be changed mid-course. [C6]

Before recruitment, one correction could be to reduce the initial task's scope and reassign existing duties to protect supervision. If no capacity can be provided, withdraw the proposed design before asking somebody to rely on it. During work, a missed review needs a real replacement or a safe change in task, not merely a new “completed” field. An unsafe request needs a reachable person with actual authority to stop or redirect it, not an AI reassurance.

A failed arrangement can leave unpaid costs, lost confidence or a foregone alternative. A later good role does not automatically restore those losses. Conversely, stopping an unsuitable trial need not end the person's future. References, an agreed work sample, a different training route or another employer may support a new prospect where the person wants one. Those are possibilities to establish, not promises this casebook can make.

What the released book contributes—and what it cannot supply

ME1.0 already discusses usable routes (Chapter 5), clocks and controlled delay (6), authored comparison (8), safe witnessing and answerability (9), residue (10), formation, shared capability and maintenance (11), and distributed responsibility (12). Applying those distinctions together keeps the beginner's future connected to the actual work of supporting it. They are existing book content, not new primitives discovered here.

What remains **UNKNOWN** is whether a real employer has an appropriate vacancy, whether the role can be funded and maintained, whether an adult wants it, whether the task teaches transferable competence, and whether our account adds anything to stronger professional practice. No independent human field encounter is claimed. What has changed in the reading is narrower and concrete: we no longer mistake training funding or another review form for the material capacity to support a first paid step.

Primary sources and correction record

- **[C1]** UK Government, Digital and Technologies Jobs Plan, 21 October 2026; foreword and Parts 2–3. Policy commitments/ambition, not observed delivery.
- **[C2]** DSIT/LinkedIn, A snapshot of entry-level hiring in the UK, 8 June 2026; Key messages. Platform-based analysis, no established AI causation.
- **[C3]** techUK, Early Careers Jobs Alliance, 11 June 2026; activity areas. Institutional description, not programme evaluation.
- **[C4]** Department for Work and Pensions, Apprenticeship funding rules 2026–27, version 3, July 2026; paragraphs 66.1, 100–103, 107.6, 112.1 and 112.11. England; relevant start-date and funding conditions apply. The official landing page was checked for current version.
- **[C5]** Claude Code's internal AI project premise challenge, 26 October 2026. Editorial criticism, not an employment source or field participant. Build independently reacquired the relevant primary rules before using the correction.
- **[C6]** Institute for Government, Doing them Justice: Lessons from four cases of policy implementation, 2014; Lesson 2. A practitioner case report used as one comparator, not evidence about this employer.
- **Book basis:** exact released v1.0.0 manuscript.

All external sources retrieved 26 October 2026. This case was authored by Framework Build with AI assistance; it is an illustrative supplement, not part of the frozen v1.0.0 release or a new licence grant.

Return to the casebook

Case 4. GitLab's 2017 database outage: what makes a recovery route usable?

Historical technical case, 31 January–10 February 2017; source and editorial review updated 27 October 2026. An online casebook addition, **not part of the dated 26 October three-case PDF**. No user testimony, service testing or independent operator audit was conducted. Existing reliability/SRE methods are the first comparator, not competitors to dismiss.

Historical operator reports are distinguished below from subsequent tracking statements and our engineering analysis. The key question is whether a stored copy can become **usable and reconcilable service** while retaining what the copy cannot restore.

1. The decision under examination

The central engineering decision is which recovery route to pursue after expected protection has failed. Selecting the newest available copy is an incomplete rule. A copy also has to be readable, transferable, usable by the restoration tools, and reconcilable with the application that will resume writing. Its destination must become operational before the route provides service. These requirements make recovery a problem of feasible transitions rather than a count of nominal safeguards.

The GitLab incident supplies a concrete instance. The operator's 10 February account describes accidental removal on the primary while repairing replication, unavailable expected recovery paths, and recovery using an older staging copy. Its account separates preservation of Git repositories from loss of database changes. [S1] The 1 February account is an earlier report from the same organisation, not independent corroboration. It describes interrupted service, incomplete recovery and measures taken while the incident was unfolding. [S2]

This analysis asks a narrower question than whether the organisation became reliable afterwards: what made one recovery route preferable and what did completing it leave unresolved? The evidence can support an attributed mechanism study without proving every operator diagnosis or the effectiveness of later reforms. There are no command logs, independently timed transfers, database images or affected-user interviews in this record set.

2. Evidence inventory and clock boundaries

The following inventory compresses the two operator reports. It describes reported availability during recovery, not a present-day assessment of GitLab's infrastructure. “Unavailable” means unavailable for the specified recovery purpose; it does not mean every component of that system failed.

Candidate copy or route	Reported age, purpose and location	Prerequisite or obstruction	Recovery consequence
Primary and standby [S2]	Live production and failover	Standby cleared for rebuild; primary removed mistakenly	Neither supplied the needed restore
Scheduled logical dumps [S2]	Daily export to S3	No usable recent dump; tool-version failure	Expected route unavailable
Cloud disk snapshots [S2]	Protection used elsewhere	Not enabled on database hosts	No database recovery route
Routine LVM copy [S1]	Almost a day old; staging refresh	Transfer and restoration	Larger data gap
Manual LVM copy [S1]	About six hours old; staging work	Staging storage limited transfer; webhooks needed reconciliation	Selected; sequence values also advanced

Sources: [S1], recovery sections; [S2], incident and problems sections. The destructive command was reportedly stopped within seconds. [S2] That short intervention window contrasts with a much longer restoration route; it is not a measured response-time requirement. Missing changes after the selected recovery point remained outside what loading that copy could recover.

The operator's accounts contain different endpoints. They should remain visible rather than being averaged into an apparently precise chronology.

Report passage	Endpoint actually stated	Meaning retained here
1 February update	17:20–23:25 UTC, 31 January	Early account's lost-change interval
1 February update	18:14 UTC, 1 February	Reported return online
10 February introduction	17:20–00:00 UTC	Later introduction's lost-change interval
10 February impact section	17:20–23:30 UTC	Later impact section's lost-change interval
10 February recovery section	17:00, then approximately 18:00 UTC, 1 February	Database restoration without webhooks, then final procedures

Sources: [S2], opening update; [S1], introduction, impact and recovery sections. These are stated endpoints, not interchangeable measurements. The midnight wording is reproduced without repairing its calendar ambiguity. The early account also places stopping the destructive action at 23:27; that is a different event from a declared loss boundary. [S2]

An outage clock, a recovery-point clock and a user-work clock answer different questions. An availability interval measures when a specified service could be used. A restored-state timestamp describes the age of the selected state. A lost-change interval describes changes absent from that state. Subtracting one pair of endpoints does not establish the others. Exact timestamps would not remove this conceptual distinction.

3. A feasible-route model, without a fictional optimum

Consider the choice between the two LVM routes in the inventory. This is the actual decision the model clarifies. Define a candidate route as a source copy, a destination, the operations needed to restore it, and the reconciliation needed before that destination accepts ordinary work. The model compares routes, not storage products.

Do not confuse three gates. A route that might physically work, a route whose capability has been established well enough to rely on, and a route somebody may legitimately attempt are not the same set.

1. **Physical candidate feasibility:** the source copy is in fact readable, necessary tools and destination capacity can perform the transfer, and required application-state reconciliation can be achieved. A real route may be feasible before anyone has tested it. Conversely, confidence or permission cannot make a physically impossible route work.
2. **Evidential sufficiency for a particular decision:** which of those prerequisites was observed, reported, inferred, failed or untested? Which time and workload assumptions were actually checked? When the answer is unknown, retain UNKNOWN rather than placing the route inside or outside a known feasible set by decree.
3. **Authority and uncertainty acceptance:** who can authorise the attempt, what consequences could reach other users, what can be staged or reversed, and who is entitled to accept the residual risk? An authorised attempt is not proof it will succeed; a technically feasible route is not automatically authorised.

These are analytical distinctions, **not a claim that GitLab used this three-gate model**. The evidence from [S1] and [S2] supports an operator-reported preference among surviving copies; it does not show that all prerequisites were independently verified or that risk acceptance guaranteed physical feasibility. The smallest honest decision record names the route, what is established about it, and what remains to be decided by an authorised actor.

Compare feasible routes on three separate dimensions: expected missing work, expected time to a usable service, and uncertainty about successful completion. Do not combine them into an unexplained weighted score. A score would require an authorised valuation of delay against missing work, plus estimates that these sources do not supply. Keeping the dimensions separate reveals the tradeoff instead of concealing it in arithmetic.

Comparison for this decision	More recent manual copy	Older routine copy
State age	Favours this route	Disfavours this route
Transfer and preparation duration	No defensible complete forecast obtained	No defensible complete forecast obtained
Successful restoration probability	No numerical estimate obtained	No numerical estimate obtained
Application reconciliation	Must be established	Must be established

Comparison for this decision	More recent manual copy	Older routine copy
Overall quantitative optimum	Not calculable from this evidence	Not calculable from this evidence

The age comparison supports a reason to prefer the newer copy if both routes are otherwise workable. It does not prove that preference minimised total damage. A much slower or much less credible route could reverse the choice; the necessary comparison data are absent. No hypothetical reversal is being presented as something the operator overlooked.

The model's principal assumption is that each listed copy can be evaluated without destroying the other options. If evaluation consumes the only remaining source, the decision changes: information gathering itself can reduce recoverability. A prudent procedure would preserve the original where feasible and perform restoration against a separate target. Whether every historical operation satisfied that condition is unknown here.

A second assumption is that missing changes cannot simply be recovered through an unexamined source. If another journal, client cache or independent record were available, it could alter the feasible set. The model does not declare such a source nonexistent merely because the reviewed accounts do not establish it. Its conclusion is conditional: the newer surviving copy offers the smaller apparent state-age gap among the documented alternatives.

4. Location and purpose become recovery constraints

A backup's location is often treated as an inventory attribute. During recovery it becomes part of the execution path. Source read speed, intermediate storage, destination writes and preparation steps can dominate elapsed time even when the copy itself is sound. A nominally available object therefore does not imply promptly available service.

The later account reports approximately 18 hours copying, attributing the bottleneck to staging's network disks, limited to about 60 Mbps, rather than CPU or network capacity. [S1] These are operator-reported measurements and attribution. Multiplying the rate by elapsed time would not independently validate the diagnosis: the record does not supply the sustained-rate trace, transfer overhead or exact copied byte count required for that calculation.

This is not an argument for selecting an expensive storage tier without measurement. Before choosing a design, an engineer would need representative restoration measurements at the relevant database size and under the relevant throttling conditions. A small-file copy test might establish connectivity while missing the limiting behaviour of a large restore. Conversely, a slow observation during an incident does not identify the performance of every alternative architecture.

The same distinction applies to purpose. Development copies may be intentionally altered to prevent interactions with production. That modification can be correct for development and inconvenient for disaster recovery. The engineering question is which transformations separate the source from the required target, and whether those transformations can be reversed safely. Calling the copy “a backup” does not answer it.

The useful deliverable is a restoration dependency map: preserved source, required transformations, destination prerequisites and evidence for completion. Its advantage over an asset list is falsifiability. An asset list can remain accurate while every practical recovery route is blocked. A dependency map can be challenged at the specific step where the route stops working.

5. Restoration is also reconciliation

Starting a database establishes that a process can read some state. It does not establish that the application can safely resume all behaviours associated with that state. Restoration needs an explicit boundary between data loading, application validation and renewed external effects.

The historical account reports two concrete reconciliation operations rather than only theoretical risks. Because the staging-copy procedure removed webhooks to prevent accidental triggering during development, GitLab reports restoring webhook configuration from a separate database assembled from the retained snapshot; the main database was reportedly available without webhooks by around 17:00 UTC on 1 February, with final webhook procedures finished around 18:00 [S1]. The published recovery plan also calls for increasing database sequence values by 100,000 to avoid reusing identifiers that may have been exposed before the outage [S1]. This is a reported plan/intervention in the operator account, **not proof that all external actions or client references were reconciled**.

These actions expose two general recovery questions. First, a restored webhook configuration can refer to an event that already happened outside the database, an event that should still happen, or a call that must not be repeated. Those cases require different decisions. Neither [S1] nor [S2] supplies an independently verified ledger of external deliveries, so no actual duplicate effect is inferred here. Second, an older database may forget identifiers already exposed to clients or external systems. Increasing sequence values is one protective technique reported in this incident; it does not alone establish everything remembered by those external consumers.

The lesson is not that reconciliation was absent: **it was explicitly attempted and partially described**. The unresolved question is the evidence necessary to conclude that each relevant application and external effect was safe to resume.

A restoration acceptance procedure would consequently separate at least three questions: can the database serve the application, are relationships and identifiers safe for renewed writes, and can externally effective operations resume? An affirmative

answer to the first should not silently answer the other two. This is an analytical acceptance structure; it is not a certification that the historical recovery was incomplete or unsafe.

There is also a distinction between reconstructing an object and reconstructing its history. A replacement project entry may restore access to surviving material without reproducing missing discussion, ordering, permissions or decisions. Even where some reconstruction is technically possible, its evidential status matters: a reconstructed entry should not acquire the authority of an original event merely because it occupies the same interface position.

These considerations do not justify withholding service indefinitely. They justify staged acceptance conditions. Read access, new writes and outbound effects may require different evidence. If staged restoration is chosen, the operator should state which capabilities are available and which remain constrained, so that users do not interpret one successful interaction as evidence of complete restoration.

6. Destructive repair and the shrinking option set

The destructive-operation window in the inventory is much shorter than the recovery interval. That asymmetry changes where safeguards are useful. A warning noticed after destructive execution can have limited protective value even if the operator reacts promptly. Safeguards placed before execution can act while the source still exists.

Host differentiation is one candidate safeguard, not a complete solution. A visible label helps only if it is noticed, understood and connected to the command's target. A stronger design might verify the target independently, restrict destructive operations, require a preserved recovery point or separate the relevant privileges. Each measure also has a failure mode and operational cost. This evidence does not compare their performance or authorise a universal two-person requirement.

The relevant failure class is a repair action that removes an option before the replacement is established. It appears in many domains, but that resemblance does not establish a shared causal model. Here it yields a concrete question: what remains recoverable after each step if the next step fails? A procedure that answers only how to complete the intended sequence omits its own interrupted states.

Diagnosis can also become consequential before an explicit deletion. Silence from a tool can mean waiting, blocking, failure or insufficient instrumentation. Treating those states as equivalent invites corrective actions against the wrong condition. The operational improvement is to distinguish them with evidence or a documented uncertainty boundary. It is not to assume every quiet process is healthy.

This analysis does not infer an individual's competence, motivation or emotional state. It identifies properties a repair procedure can expose to testing: target discrimination, visibility of progress, preservation of alternatives and behaviour after interruption. Responsibility for these properties is broader than responsibility for pressing a particular key.

7. What changed afterwards: bounded evidence

The original restore-testing issue proposes recovering a recent backup, examining meaningful application tables, and reporting failures through another channel. It also proposes using the same underlying tooling as manual recovery. [S3] The separate durability-owner issue proposes explicit responsibility and authority to interrupt work when durability is endangered. [S4] These descriptions establish proposals, not demonstrated outcomes.

Additional reading changes the scout's evidential ceiling. Tracking issue 1684 reports several completed measures and describes transitions between proposed solutions. It also distinguishes streaming-backup implementation from missing monitoring and carries unfinished restore-related work forward. [S5] Streaming issue 1152 has checked items for backup use, configuration tooling and documentation; its rationale couples backup restoration to development use. [S6] Issue 1265's retrieved description retains unchecked host-creation, configuration and restoration tasks. [S7] The hourly-snapshot issue states a proposed frequency and retention change. [S8]

The earned result is therefore narrower than “all improvements unknown” but weaker than “recovery proved reliable.” There is public operator-maintained evidence reporting some implementation. There is no retrieved execution history or independent test record establishing end-to-end recovery capability, and the visible descriptions do not provide a reliable completion chronology. A checked box can identify a claim to investigate; it cannot substitute for the result of a restore under specified conditions.

Coupling restoration to development has a plausible mechanism: recurring useful work creates opportunities to discover broken tools and incompatibilities before an emergency. It also has a possible blind spot. A development restore may omit production traffic, effect handling, privilege boundaries or the destination constraints that matter in an actual incident. Routine reuse improves the relevance of a test only to the extent that the exercised path overlaps the required recovery path.

An owner can likewise improve escalation without controlling every dependency. The necessary evidence would connect responsibility to an observed ability to stop, repair or resource the failing route. Merely naming an owner would not establish that ability. Neither limitation makes the proposals worthless; both specify what would distinguish a paper control from a working one.

8. Reliability objectives and the remaining consequences

Ordinary database reliability supplies the strongest comparator. A recovery point objective specifies a target for acceptable data loss or recovery-point age. A recovery time objective specifies a target duration for restoring a defined service. These are objectives, not descriptions of whatever happened. The reviewed material does not establish the adopted objectives needed to judge compliance quantitatively.

Even an objective met exactly would leave questions about distribution. A bounded interval of missing state can contain very different amounts of work for different users. Duration is not a measure of effort, replaceability or consequence. Those differences do not require an invented harm score; they require evidence about affected activities before making claims about them.

Service restoration is a real result and should not be rhetorically erased. It makes further work possible. At the same time, restored access does not imply recovered discussion, repeated effort or reversed downstream decisions. This draft has no user-level evidence with which to estimate those consequences, and makes no allegation about particular users' subsequent outcomes.

Mechanical Ethics can direct attention to that separation, but database reliability, human factors and incident management already support much of the analysis. Nothing here demonstrates theoretical novelty or an advantage from applying ME. The case's contribution is a specific, inspectable recovery decision and its residual uncertainties, not a claim that familiar engineering has been newly explained.

The defensible conclusion is conditional and operational: evaluate recovery as a route through preserved state, executable dependencies and application reconciliation. Record what becomes usable, what remains missing and what evidence supports each claim. Repeatedly exercised recovery can strengthen that route; an inventory, a returned homepage or a completed issue alone cannot establish its full capability.

9. How to falsify the extra Mechanical Ethics contribution

Existing database-reliability practice already distinguishes recovery point and time objectives, verified restores, application consistency, accountable postmortems and safer deployment. This case does **not** establish that applying Mechanical Ethics led GitLab to recover faster or differently. In a future comparable real-world recovery exercise, the stronger comparison is with an experienced incident team using its ordinary SRE procedure. Before adding ME terminology, record which hidden dependencies, external effects, possible losses, affected parties and decision clocks the team already identified. Then ask whether the extra boundary/burden/residue questions uncover anything missing and consequential. **EXISTING_METHOD_BETTER** and **NO_MATERIAL_DIFFERENCE** are permissible results. An attractive model with more fields and no changed finding may impose cost without benefit.

The case's positive construction is the restoration of service and the subsequent work to create a **maintained, testable recovery capability**. That capability needs ordinary practice, ownership, exercises, tools, people and budgets; an after-action checklist or completed issue is not the same as a tested recovery path. Rebuilding these routes matters even though the missed database changes from this particular incident cannot be put back by a later successful restore.

Source key

- **S1:** GitLab, 10 February 2017, Postmortem of database outage of January 31.
- **S2:** GitLab, 1 February 2017, GitLab.com database incident.
- **S3:** Original infrastructure issue 1102, automated recovery testing.
- **S4:** Original infrastructure issue 1163, durability ownership.
- **S5:** Original infrastructure issue 1684, incident follow-up tracker.
- **S6:** Original infrastructure issue 1152, streaming backup.
- **S7:** Original infrastructure issue 1265, automated restoration.
- **S8:** Original infrastructure issue 1098, hourly snapshots.

Case 5. The National Library Service: building and maintaining a usable reading route

Historical and continuing constructive public-service study; United States, 1931–2026. Prepared 27 October 2026 from named public records. This is an **online addition after the dated 26 October three-case PDF**, not an amendment to the frozen Mechanical Ethics v1.0.0 book, an accessibility evaluation by patrons, or a determination of anyone's legal entitlement.

1. Begin with what becomes possible

An accessible book is not simply a text converted to another format. Somebody must select or produce it, make it discoverable, lend it to an eligible reader, supply a way to play or read it, offer support if the device fails, and keep the arrangement working as formats and people's circumstances change. A public library service can form a durable shared capability: a person can read materials that would otherwise be inaccessible through standard print, without having to purchase a bespoke library of their own.

The U.S. Library of Congress's National Library Service for the Blind and Print Disabled (NLS) provides an actual long-running example. Established under federal legislation in 1931, it lends audio and braille materials and the equipment needed to use them without charging eligible patrons. Materials can travel digitally through BARD or physically through a network of cooperating libraries and postage-free mail. The Library describes a national network coordinated across federal, state and local levels. [N1, N2] This is a real service and an important positive achievement; the case does not attribute its creation to Mechanical Ethics.

Our question is narrower: **what joins a formally available right or service to an actually usable and maintained capability, and what changed when independent scrutiny revealed an access barrier?**

2. Boundary, affected people and evidence routes

The analysis follows a prospective reader's journey from becoming aware of NLS to receiving readable material. It also follows two changes: the 2016 amendment permitting equipment for electronic braille, and the 2021 expansion of professionals able to certify eligibility. [G1, R1]

The relevant entities have different powers and dependencies:

Role	Contributes to reading access	Cannot by itself establish
Congress and the Library of Congress	Statutory scope, funding and programme direction	That a particular reader knows how to use the service

Role	Contributes to reading access	Cannot by itself establish
NLS headquarters	Accessible materials, device policy, production and technology	That every applicant can complete local registration
State/local network libraries	Applications, enrolment, distribution, reader advice and equipment support	That federal format policy matches every patron's situation
USPS and delivery network	Postage-free movement of physical media and devices	That downloaded material or electronic equipment is suitable
Eligibility certifiers	A permitted route for establishing qualification	That a certified patron will have useful devices, materials or support
Potential patrons and patrons	Needs, reading choices and evidence of practical use	The national eligibility denominator or programme-wide outcomes
GAO and other reviewers	Independent challenge and documented recommendations	That a formal recommendation marked implemented caused increased reading

These are not interchangeable participants. A central agency may have the power to broaden the certification rules while local libraries carry much of the work of applications, circulation, equipment repairs and direct advice. The NLS account says state, local and/or federal sources fund network libraries; one central appropriation therefore should not be passed off as the programme's entire social and operating cost. [N1, N2]

This case does not interview or collect data about individual disabilities, reading preferences or rejected applications. The gap between a rule and lived access remains a **question for affected readers and domain owners**, not a story we invent on their behalf.

3. The institutional barrier GAO actually observed

GAO's 4 April 2016 audit examined NLS's services and use of changing technology. It recorded two concrete limits. First, someone seeking service for a reading disability such as dyslexia generally faced a **medical-doctor certification condition** that applicants with other qualifying disabilities did not face. GAO recorded concerns from network library officials and others that the extra steps and costs obstructed access. GAO interviewed eight selected network libraries; their experience is material evidence but not a nationally representative measurement of all eligible people. [G1]

Second, the statute then constrained using NLS funds to provide equipment for electronic braille, even though NLS already made electronic braille material available. A file could exist while the funded route to a necessary reading device remained outside the programme's authority. The solution could not consist solely of teaching librarians to be more helpful: one missing capability was a legislative permission to purchase and lend a different class of equipment. [G1]

Notice how these two barriers operate at different points. The certification rule concerns **entry**; the equipment rule concerns **use after entry**. A single headline such as 'accessible library exists' loses both distinctions. A bare count of registered people also cannot reveal everyone deterred before registration.

4. Two corrections and two clocks

Event	What is established by the named source	What does not follow
April 2016: GAO report	Independent federal audit reports certification and device-authorisation issues; recommendations and a matter for Congress [G1].	Every eligible reader was counted or personally interviewed.
July 2016: Public Law 114-219	Congress amended the equipment authorization; GAO's status page describes the matter as implemented [G2].	Actual distribution or individual use followed immediately.
February 2021: certification rule	The final rule expanded permitted certifiers to include educators, psychologists, certain nurses and other qualifying professionals, and removed the special medical-doctor requirement for reading disabilities [R1].	Previously excluded applicants all enrolled or their experience was measured.
2024: braille-device rollout	NLS reports nationwide distribution to network libraries with staff training and approximately **8,500** eReaders with patrons and network libraries [N3].	Those are 8,500 unique newly served people or an effect size caused by GAO.
FY2025 reported in 2026	NLS says nearly **17,000** of its new DA2 talking-book players had been distributed and a separate VA sign-up collaboration added nearly **3,000** patrons [N4].	These figures quantify the effect of the 2021 certification change or service for all eligible people.

The equipment authority changed within months of the audit. The certification rule changed years later. The evidence does not support a single 'the programme was fixed' timestamp. There are at least four distinct clocks: legislation, rulemaking, physical capability distribution and individual practical access. The first two can be observed in published acts and rules. The last needs patron-level evidence that the published records here do not supply.

GAO's status page marks its five listed congressional/executive recommendations as **closed/implemented** and describes follow-up studies, an eligibility pilot and other administrative actions. [G2] That is meaningful evidence of specified follow-up activity and legal change. It is not an independent field measurement of how many reading opportunities were added or whose access failed afterwards.

5. The complete usable route is composed, not declared

For a particular reader and item at a particular time, a successful route might require:

1. **Discover:** the reader learns the service exists through a route they can actually use.
2. **Qualify:** the person meets the applicable eligibility conditions.
3. **Certify:** someone authorized and accessible can complete the relevant certification without excessive delay or cost.
4. **Enrol:** the network library processes the application and tells the patron the next step.
5. **Obtain:** the desired item exists in an appropriate medium and can be delivered or downloaded.
6. **Use:** the patron has appropriate equipment, training or assistance to read the item.
7. **Continue:** equipment, repairs, software, staff support and continued lending survive over time.

These are analytical dependencies, not a checklist NLS claims to have adopted. They expose multiple useful intervention points. Improving certification can remove an entry barrier, while a missing device can still block practical use. Providing a device can widen possibilities, while an online-only route could impose a new burden on someone without reliable internet. The programme's combination of physical lending and downloads is therefore relevant: new technological options need not erase older useful options. [N1, N2, N3]

Formally, for one declared reading goal, we can denote the composed route as a conjunction of necessary usable transitions. Missing evidence for one necessary step yields **unestablished completion**, not a verdict that all service has failed. If a helpful neighbour, network librarian or alternate format establishes a different route, the model must represent it rather than infer none exist.

Neither a federal rule nor a device shipment certifies end-user usability. The local network and the user's own account are necessary additional apertures. Equally, a measured service benefit is not disproved by a missing evaluation of one subgroup.

6. Who carries the continuing work?

NLS is a constructive case, rather than just a failure-and-appeal story. Its persistent capability depends on continuing contributions: choosing and producing accessible books, making delivery systems work, repairing machines, answering questions, certifying applicants and renewing formats. The service is valuable because people perform those functions repeatedly, not because an act was passed once in 1931. [N1, N2]

The burden ledger is consequently plural.

- NLS carries central content, policy and device-development work.

- Local network libraries carry registration, circulation, equipment loans and reader-support work, funded from a combination of government sources.
- USPS carries a separate physical distribution function.
- Potential and actual patrons can still carry effort in finding the service, navigating eligibility, learning technology or reporting failure.

The fourth item is an analytical possibility, **not** a quantified burden study of patrons. The first three are described by the operators' public records. Without an audited total of central, network, postal and unpaid work, it would be misleading to report a single 'cost per reading opportunity' from NLS's central budget alone.

Maintenance may also create new choices. A local library that lends hardware with support can help a patron use a new format without requiring that patron to master every implementation detail or transfer control of their reading history to a platform. Whether a particular patron prefers downloaded, posted, digital audio or braille material remains that patron's question.

7. Reading the positive results without inventing causation

The strongest **observed** positive result in this bounded source set is not a modelled saving. NLS reports an operating national lending service and actual nationwide braille eReader distribution with staff training. GAO's identified legal barrier to lending that equipment was subsequently changed. [G1, G2, N1, N3]

That sequence supports a possibility-enabling institutional mechanism: change the authorisation, deliver equipment and maintain the network that can make it usable. It does **not** establish that the change caused an independently estimated increase in literacy, enjoyment, autonomy, patron health or enrolment. None of those outcomes was measured by this case.

The 2025 reported VA sign-up and player distribution demonstrate additional activity and access routes [N4], but those are operator reports about distinct initiatives. They are not independent replications, and they do not directly quantify the effect of the 2021 rule. Likewise, fewer patrons in one year could reflect many competing explanations; we have not sourced a consistent population denominator or eligibility breakdown for a causal trend analysis.

One protected negative is important: the **public source extracts used here do not supply a defensible count of everybody eligible for the service**. GAO's 2016 public summary discusses users it observed and the need to evaluate outreach; it does not provide a reliable population denominator for this chapter. [G1] Without such a denominator, nationwide coverage rates cannot be derived from patron totals alone. Applications and circulation data may still improve a service without proving every potential reader was reached.

8. Strongest ordinary method, proposed comparative use

This case already has a competent and stronger domain method: GAO's programme evaluation, legislative change processes, accessible-service standards and network-library practice. These processes found the concrete barriers without Mechanical Ethics. We therefore **must not** credit the framework with either discovery or repair.

A meaningful future comparison could give an accessibility professional and a consenting eligible reader the ordinary NLS evaluation route first. Record how they already assess application certification, multiple formats, device availability, service maintenance and the outcome meaningful to that reader. Then use ME's five questions to see whether it adds an overlooked reachable correction route, an unacknowledged network cost, or a useful new choice.

This comparison can legitimately return **EXISTING_METHOD_BETTER**, **NO_MATERIAL_DIFFERENCE**, **USEFUL_ADDITION** or **INDETERMINATE**. It would require participant consent, independent assessment, careful handling of personal information and no implication that sharing testimony is required to receive assistance. The present chapter reports **no participant** and no such finding.

9. What this case contributes — and what stays open

The NLS example broadens the online casebook beyond institutions failing to protect people in time. It shows a public capability built and maintained for decades, independent audit identifying specific barriers, genuine legislative and regulatory corrections, and operator-reported distribution of new means of access.

A practical distinction survives: **opening a legal door, delivering a tool, and a person actually being able to use it are different transitions**. These transitions can be joined constructively, with costs and authority distributed across organisations. But the records used here cannot tell us how a particular person's life changed after the reforms, or whether ME is better than existing accessibility practice at seeing the relevant steps.

An independent, affected reader's experience and strong specialist method remain the next owners of those questions. No moral standing, entitlement, eligibility outcome or advice about a present application is decided by this account.

Sources and scope

- **[N1]** Library of Congress, NLS overview (current operator description, retrieved 27 Oct 2026); national service, 1931 history, formats, distribution.
- **[N2]** Library of Congress, Network Libraries (operator description, retrieved 27 Oct 2026); division of network roles, funding and support.
- **[G1]** U.S. Government Accountability Office, GAO-16-355, 4 April 2016 (HTML executive findings and recommendations, checked 27 Oct 2026); access barriers, technology authority, 2014 user limitations and review scope.

- **[G2]** GAO-16-355 recommendation and congressional-matter status on the same page (checked 27 Oct 2026); a follow-up status page, **not independent of [G1]**.
- **[R1]** Library of Congress, Federal Register final rule, 86 FR 9289, 12 February 2021 (official-agency text in the Federal Register's accessible HTML rendition, checked 27 Oct 2026; the site notes the govinfo PDF is the official legal edition). The rule's effective date and revised certifier list, not evidence of actual patron registration.
- **[N3]** NLS, News October–December 2024 (operator self-report of nationwide eReader rollout, training and approximately 8,500 devices).
- **[N4]** NLS, News January–March 2026: 2025 in review (operator self-report of DA2 devices, VA enrolment collaboration and programmes; no attributable before/after evaluation).

Limits: The official public HTML reports and named online text were examined; no personal application records, local network interviews, full GAO PDF appendix, rule-implementation logs, independent device-usability trial, nationally consistent eligibility denominator or patron-level before/after data were read or generated. The central findings are source-relative. This chapter does not validate NLS's aggregate performance, establish why readership changed or measure the effect of applying Mechanical Ethics.

Comparative analysis

Scope

The two historical cases and the employment design study have different units of analysis. The disclosure case follows requests and records through an institutional process. The fire case follows people and protective operations through a changing physical environment. The employment study examines a proposed allocation of work, supervision and funding. They are not independent replications of one intervention.

Common structure and domain-specific criteria

Analytical distinction	Information disclosure	Wildfire protective action	Employment design
Input	Request and identifying information	Hazard information and warning	Funding, candidate and task
Intermediate operation	Preprocessing and record matching	Reception, mobilisation and movement	Instruction, practice and review
Required conditions	Applicable disclosure conditions and recipient safeguards	Usable route or refuge under relevant hazard conditions	Pay, competent supervision, access and task suitability
Output	Released record	Reached location or completed rescue	Completed work and assessment
Outcome requiring separate evidence	Lawful use and maintained confidentiality	Survival, injury and subsequent exposure	Learning, retention and sustainable productive capacity
Correction after initial failure	Restrict further disclosure or later use	Reroute, shelter or rescue	Reallocate work, supervision or funding
Residue	Information already disclosed	Injury, loss and displacement	Lost time, income or confidence

The common structure is a sequence of conditional transitions. Evidence that one transition occurred does not establish that later conditions were met. However, the conditions themselves are not interchangeable. A matching threshold cannot determine a statutory entitlement. A fire-arrival estimate cannot establish an individual's mobility. A balanced funding allocation cannot establish learning.

Failure classification

Three failure classes should be distinguished.

Specification failure: the selected rule or objective omits a material condition. A perfectly implemented format check can still be insufficient for a substantive disclosure requirement. A plan that specifies warning issuance but omits transport capacity can be complete against its own specification and incomplete against the intended protective effect.

Execution failure: an adequate requirement is not realised in operation. Examples to investigate include a matching branch bypassing a required check, a warning not reaching its intended recipient, or scheduled supervisory hours being displaced. Whether a particular event belongs here requires evidence about both the requirement and its implementation.

Model failure: the account used to assess adequacy misrepresents the system. Examples include treating available road capacity as independent of simultaneous demand, assuming that all supervisory hours qualify for training funding, or using a retrospective outcome category as a prospective causal estimate.

These classes can coexist. Correcting one does not demonstrate correction of the others. Their practical value is to identify the evidence and authority needed for the next decision, rather than to assign a general failure label.

Causal identification

The disclosure record supports analysis of documented processing and legal findings. It does not identify a population-level effect on filing behaviour. Declarations from affected organisations are relevant observations, but they cannot alone estimate the counterfactual number of returns that would otherwise have been filed.

The fire research supports retrospective reconstruction and differentiated outcome classification. It does not provide a randomly or otherwise comparably assigned contrast between refuge policies. People who entered refuge areas were selected by exposure, route conditions, responder decisions and mobility. An unadjusted comparison with people who evacuated would combine those differences with the effect under investigation.

The employment study has no observed outcome to explain. Its calculations are feasibility conditions. They can identify an inconsistent proposal; satisfying them cannot establish that an actual arrangement will succeed.

Consequently, the studies do not support an aggregate estimate of the effect of using Mechanical Ethics. Such an estimate would require a specified intervention, comparison condition, outcome definition and observation protocol. This does not prevent structural analysis; it limits what follows from it.

Correction and verification

A corrective action requires an owner able to change the relevant condition, an observable completion criterion and a means of detecting residual failure. The owner of a diagnosis need not own the remedy. A technical team may expose a branch defect while courts determine legal restrictions; researchers may expose a refuge-planning problem while local authorities control implementation; an employer may expose a recurring funding constraint while the rule owner controls policy revision.

Verification should examine the claimed effect. A revised procedure is not evidence that previously released data were recovered. A designated site is not evidence that it remains accessible during a fire. A funded training plan is not evidence that productive supervision is available. These are distinct empirical questions with different observation requirements.

Contribution and limits

The main work supplies a vocabulary for retaining these distinctions across institutional boundaries. The case studies supply domain detail and expose where a compressed description would be misleading. Neither replaces the other. The strongest contribution demonstrated here is an inspectable analytical structure; improved decisions or reduced harm in practice remain unmeasured.

Contents

Reading across five cases: routes that fail and capabilities that form

27 October 2026 — online addendum, NOT part of the dated 26 October three-case PDF or portable HTML reader. Read the original three-case comparative analysis first if you want its intact initial scope. This addendum adds two later historical source studies, GitLab's 2017 recovery and the National Library Service, without rewriting the original studies or their published source snapshot.

What changes when the casebook includes a constructive public service?

The first three cases were selected to examine disclosure, emergency protection and viable supervised employment. They are not replications: each has different facts, forms of authority and specialised methods. Case 4 adds an IT recovery path where multiple nominal backups failed and partial restoration could not recreate missing work. Case 5 adds a long-lived library service that makes accessible reading possible through joined institutions and shows specific documented changes to previously restrictive access routes.

A framework that only says 'harm is not repaired by closing a file' can illuminate a failure and still leave the useful next step unspecified. A constructive case asks a different question: **which complementary capabilities have to exist, who keeps them working, who may refuse, and what improvements are actually observed?**

Case	Strongest existing domain methods	Narrow ME reading that may help	Supported actual event/result	Not established
1. IRS–ICE disclosure	Tax confidentiality law, applicable court review, agency audits	Substantive legal conditions vs checks that could detect failure; information once disclosed vs later restrictions	Attributed records of disclosure and preliminary court findings, dated to original study's September 2026 cutoff	Final merits, downstream address use, improved outcomes from ME
2. Camp Fire evacuation	NIST fire research, emergency management, route and tenability models	Segment-specific feasible protective journey, not merely a warning or roadmap	Retrospective event research and differentiated refuge/outcome categories	A prospective refuge-policy effect, causal lives-saved estimate, safe-site certification
3. Paid first technical rung	Employment/ apprenticeship rules, supervision design, professional staff budgeting	Funded or advertised work vs competent paid support and meaningful learning	**Constructed** necessary capacity conditions against real rules	An observed employer, successful hire, participant learning or incremental ME benefit

Case	Strongest existing domain methods	Narrow ME reading that may help	Supported actual event/result	Not established
4. GitLab recovery	Database operations, backup restores, incident response, RPO/RTO	Physically possible recovery vs established evidence vs authorised attempt; recovered state vs external effects and missing work	Historical operator reports of staging restore, webhook reconciliation and continued lost database changes	Independently replayed incident, user-level aftereffects, all follow-up controls tested, ME advantage over SRE
5. NLS reading access	Public-service evaluation, accessibility law/ standards, network-library practice	Rule authorisation vs produced/ delivered equipment vs a reader actually using a maintained route	GAO criticism, subsequent statutory/regulatory changes and NLS-reported device distribution	Causal effect of those changes on eligible population, individual usability, incremental ME advantage over specialist methods

The shared mechanism is not one universal clock

The cases share a need to connect **an account** with a transition that produces the outcome an affected party needs. But the controlling conditions differ. A statutory disclosure test is not a physical evacuation threshold. A restoration point is not an end-user outcome; a library item's existence is not the same as being able to read it. An employment hour that can be financed is not necessarily competent support available at the right time.

Three failure families from the original comparison remain useful and can be read constructively:

- **Specification:** necessary downstream conditions were not named. A backup inventory without an executable restore path or a library programme without a usable application route can satisfy an incomplete description.
- **Execution:** named conditions did not come about in the relevant time and place. Example questions include whether recovery hardware was available at the needed throughput or an applicant actually reached an authorised certifier.
- **Model:** the description or projected effect misstates what is reachable. The most dangerous version promotes a proposal, authorisation, delivery count or administrative closure into proof that the intended capability now exists.

All three can coexist. Calling one an error does not determine who is to blame or who may legally act; those require the relevant evidence and domain process.

What can actually be built?

The two new additions provide a useful contrast.

In GitLab, recovery restores a service through a chain of surviving snapshots, storage performance, correct identifiers, webhook data, tooling and people. The historical service came back online, and some database work remained unrecovered. The next constructive task is **a routinely exercised recovery route**, not merely the elimination of a failed test result. The operator's tracked follow-up shows proposals and some reported implementation, not an independently measured guarantee.

NLS shows a capability sustained over generations. Headquarters, network libraries, equipment, postal delivery and varied certification paths connect to the reader's ability to use literature. The 2016 audit found two distinct obstacles, law about braille devices and certification requirements. Subsequent rule and statute changes and device distribution show specific positive changes, while end-user access and population coverage remain incompletely observed.

These examples resist opposite shortcuts: **a damaging incident is not a complete story of its repair; a constructive programme's longevity does not certify every participant's access.**

Cross-case falsifiers and the positive test

The framework should be compared with specialists' ordinary methods, not credited retrospectively for finding what those methods already found. A prospective evaluation should predeclare the existing method and measure actual use burden. Useful outcomes might be identifying a missing route with an available owner, avoiding a false claim that a check was capable, finding an unacknowledged maintenance cost, or preserving one person's option without inventing consent.

The following results all count as intellectually valid: **EXISTING_METHOD_BETTER**, **NO_MATERIAL_DIFFERENCE**, **MATERIAL_ADDITION** (a corroborated consequential difference) and **INDETERMINATE**. An elegant diagram with no change in a domain expert's account is not proof of practical value.

Specific counterexamples to seek:

- A recovery specialist already tracks application identifiers, external effects and restore exercises; ME adds only vocabulary. Record NO_MATERIAL_DIFFERENCE rather than renaming their work.
- A reader accesses NLS successfully, but our model incorrectly describes their preferred route as digitally unavailable or burdensome. Update the boundary, do not erase their testimony to save the theory.
- A constructive plan raises one party's capability by imposing unagreed ongoing work on another. Observe whether the 'positive' account still preserves both parties' options and claims.
- An apparently corrected rule leaves no measurable change because another access step remains blocked. That is not evidence the rule change was imaginary, nor that outcomes improved.

- A warning or corrective notice arrives after one loss but before a different loss. Label the relevant clock rather than calling the whole case either on time or too late.

None of these five existing studies measures the causal effect of introducing Mechanical Ethics. Their value is to offer **source-backed situations for such a future test**, including a maintained capability with observed institutional changes as well as late-correction cases.

Source and editorial scope

Cases 1–3 retain their original source cutoff, attribution and status. Case 4 is a new reading of GitLab's original operator reports and mutable follow-up work items, with no independent incident replay or affected-user study. Case 5 uses GAO, the Library of Congress, a final rule and NLS operator accounts, with no participant interview or causal outcome estimate. Both additional cases were source-checked on 27 October 2026, with partial archival coverage openly reported.

The 26 October 60-page PDF and its offline HTML source snapshot still contain **the original three cases only**. New online Markdown cases do not silently change those immutable readings, nor the v1.0.0 book, tags or rights.

Source receipt for Case 1 (institutional trust)

Companion to `casebook/01-institutional-trust.md`. Prepared by Claude Code on 26 October 2026, from public sources only. Each file was retrieved without credentials (`curl` or a web fetch), and its text was extracted locally with `pdftotext -layout` or an HTML-to-text step. Hashes are of the bytes as retrieved. A later download may differ where a host regenerates the page; the Cornell statute pages are dynamic HTML and are the likeliest to change.

Files read

#	Source	Bytes	SHA-256	How much was read
1a	D.C. Cir. No. 26-5006, opinion, 8 Sept 2026 (CourtListener RECAP doc 2191763)	248,919	`a54ebdf96068b8276fc873eab72f09412fa3b29973d132720c4fa5da3ce7e37b`	whole (32 pages)
1b	D.C. Cir. No. 26-5006, judgment, 8 Sept 2026 (doc 2191760)	61,626	`0b9e1d4609344f13d602bf83de301de926ecaab32a4539dc8e7dc3274acba179`	whole (1 page)
2	D.C. Cir. No. 25-5181, Centro, opinion, 24 Feb 2026	382,675	`3fac0632bb1d427f171ba4bcbf6f50bbfc6d876c29f304c73c6c1c0d4c738a5b`	**partly**: date, panel, author and the "crystal clear" holding passage. The rest of its holding is taken from CRS (source 4) and from the quotations in source 1a
3	TIGTA Report 2026-IE-R010, 4 June 2026	935,341	`49b401f5f5e7a6450f67a65d967eaa289b84f7292e6bc3983d7bc4d87251d89e`	whole, including appendices; redactions as published
4	CRS Legal Sidebar LSB11413, updated 17 Sept 2026 (EveryCRSReport mirror; congress.gov returned HTTP 403 to this fetch)	34,558	`49364b6497dcde53b8249341deec92aa331cbb431a3027572cc0a2756ab754a5`	whole

#	Source	Bytes	SHA-256	How much was read
5	Agreement and IRS declaration, ECF 30-1, No. 1:25-cv-00677 (EPI-hosted copy)	1,952,703	`9fb3ed8cf3faf9435944671511c5be6ea9dec6ddf0fbd72e4ef468b2a0763210`	every page with extractable text. Pages 5, 7, 8, 10, 16, 17 and 19 have no text beyond page headers, and pages 1 and 4 are exhibit covers. Those pages were **not** read (no OCR)
6a	26 U.S.C. §6103, Legal Information Institute page	832,960	`2617b29d61d1a5dc3efcf033c2f1671a64bd78b6d1d5273487b32fafd90eef06`	§6103(b)(6), (i)(1)(A), (i)(2)(A)-(C)
6b	26 U.S.C. §7213	65,859	`4cea041327b718a32bc35572ae807f73e28b287192f08a0e74047b27f385ae64`	(a)(1)
6c	26 U.S.C. §7431	58,165	`da04797d28c4a9247972412ea17a7261d041c6d6d92e6402bdd85290a19457da`	(c)(1)
7	Mechanical Ethics v1.0.0 master, tag `v1.0.0`	136,286	`1315a2c2b500ae2308afd7070d5b3919b23d1390870e65eea54bbc057e6e37a8`	the glossary, Notes, Ch. 6 lines 154-200, Ch. 9 in full, Ch. 10 lines 342-360, Ch. 12 lines 466-484; byte-identical to the released file

URLs are listed in the case's own source list.

Claim check

Status key:

- **P**: read in a primary source (statute, court document, agency or Inspector General record).
- **S**: read only in a secondary source (CRS) or a court's quotation of another document.
- **A**: attributed claim, reported as someone's statement, not verified.
- **U**: UNKNOWN in the sources used.

Claim in the case	Source(s)	Status
Agreement signed 7 Apr 2025; no request made by then	5 (IRS declaration ¶¶3-5); 3 (App. III)	P

Claim in the case	Source(s)	Status
Implementing agreement 18 Apr 2025	3 (App. III)	P (its text not seen)
ICE request for about 1.28 million last known addresses, 27 Jun 2025	1a (Lyons letter, Romo Decl. ¶4); 3 ("over 1.2 million"); 4	P via court record; the letter itself not seen
Earlier ICE request about more than 7 million people	1a, quoting the district court	S
47,289 records disclosed, 7 Aug 2025	1a; 3; 4	P, but every count traces to IRS records and the Romo declaration
About 46,965 individuals	3, footnote 4	P (TIGTA's own analysis of IRS data)
42,695 (90.3%) matched by taxpayer number	4; 1a ("more than 90%", "90.3%")	P via court; the declaration itself not seen
"Less than 5%" potentially incomplete or insufficient ICE addresses	3; 4	A (IRS's own estimate)
Only last known address shared, no other tax information	3	A (IRS statement to TIGTA)
Address sent undated	1a	P
Procedure checked only that fields were non-empty and that the address field had five or nine digits	1a; 3; 4	P
One point of contact for every request	1a, quoting the district court	S
Reason "more circular than specific"	1a, quoting the district court	P (appeals court adopting the phrase)
Individual review under the IRS manual before 17 Apr 2025	1a (IRM 11.3.28.2)	S (manual not read)
IRS's earlier position: no provision for immigration enforcement; address-only disclosure prohibited	1a, quoting 89 Fed. Reg. 93174 and Centro	S
Government acknowledged unlawful transfers at argument	1a (Oral Arg. Tr. 8:19-9:3, 15:17-19, 28:11-13)	S (transcript not read)
Government counsel not aware what the IRS did to fix the procedure	1a (Oral Arg. Tr. 46-48)	S
District court order 21 Nov 2025 (stay, injunction, advance notice)	1a (twice), 2, 3, 4	P; see discrepancy D1
Massachusetts order barring ICE and DHS use, 5 Feb 2026	4	S
DHS and ICE confirmed non-use pending the lawsuits	3	A (IRS statement to TIGTA)

Claim in the case	Source(s)	Status
TIGTA concerns Nov-Dec 2025 led to the IRS's manual review; IRS notified DHS 23 Jan 2026	3	P
ICE safeguard findings open; corrective plans missing or undated; review postponed	3	P for TIGTA's account; A for the IRS's statements within it
Center's clinic: "drastic" fall in cases; "nearly 10 percent" of expenses moved to outreach	1a (Olson Decl. ¶¶41-46, as found by the district court)	S
Centro: the agreement likely complies on its face; addresses are not taxpayer return information	2 (holding passage); 4; 1a	P for the passage read, S for the rest
Statute text: §6103(i)(2) four request elements, "personally and directly engaged", identity rule, (b)(6) definition, (i)(1) ex parte order	6a	P
Penalties: §7213(a)(1) felony; §7431 the greater of $1,000 per act or actual damages	6b; 6c	P
Congressional letters 12 Feb 2026; hearing 4 Mar 2026	4	S
First Circuit argument scheduled 7 Oct 2026	4	S (outcome not checked)
Whether any disclosed address was used before 5 Feb 2026	none	U
Whether affected people were notified	none	U
A population-level change in filing	none found; one standard web search found advocates' reports, no systematic 2026 data	U

Discrepancies found, and how the case treats them

- **D1, order date.** Source 1a cites "Order of Nov. 21, 2025" on its pages 9 and 31. Its conclusion and the judgment (1b) say "order of November 1, 2025". Sources 2, 3 and 4 give 21 November. The case uses 21 November and records the slip.
- **D2, requested count.** TIGTA says "over 1.2 million"; the court and CRS say about 1.28 million. These are compatible; the case gives both.
- **D3, records against individuals.** TIGTA's background text says "approximately 47,289 individuals", but its own footnote finds those records correspond to about 46,965 individuals. The case uses 47,289 records and about 46,965 people.
- **D4, a search summary's figure.** A web-search result summary gave "47,489" for a third-party case page. That page was not read, and the figure appears in no source read here. It is not used.

- **D5, a model summary's addition.** An automated summary of the CRS sidebar named the IRS's chief executive. The visible CRS text (source 4) does not: the name appears only inside the URL of its link to the hearing page. The case does not use the name.

Not used

- **The interview with Michael Lewis that prompted this case.** I did not read the transcript. No fact in the case rests on it. Where a figure attributed to it appears (seven million), the case cites the court record instead.
- **Press reports, including the Reuters article cited elsewhere in the project.** I did not read them, and no fact rests on them.
- **Sealed filings, the implementing agreement, the image-only pages of source 5, the oral-argument transcript and the Romo declaration itself.** None was read; each is cited only through a source that quotes it.

Limits

This receipt records what was read and where each claim comes from. It is not a legal opinion, and no lawyer reviewed it. The court rulings are preliminary. A reader relying on any legal point should check the current docket.

Build integration note

The tables above preserve Claude Code's original source-reading receipt. Build's editorial changes to the chapter are recorded in the whole-casebook review. In particular, the original abbreviated damages description is no longer used as a complete account: Build checked §7431(a)–(e), including liability conditions, exceptions and mandatory notification triggers. That additional reading is Build's, not a retroactive extension of CC's reported read of subsection (c)(1).

Camp Fire chapter: integration record

Codex Windows authored the standalone chapter under the supplementary casebook commission. Framework Build read the whole returned chapter, source receipts and validation record before integration. This is editorial source review, not an independent reconstruction of the fire or a practitioner review.

At first integration, `02-camp-fire.md` was byte-identical to the returned chapter: 21,449 UTF-8 bytes; SHA-256 `ca20a769e71b04283438aa7d0cd52213a25ea466437fee976f7c49768775f96d`. The public handoff is COM #877 comment 6425409312. Its private custody record is not required for readers to follow the public primary sources listed in the chapter.

The author inspected official abstracts, programme material and relevant publisher HTML sections. Full technical-report PDFs were not reviewed: tool size/content-type failures and a native HTTP 403 were retained. No PDF page inspection, site assessment, field efficacy or independent resident account is claimed. Later publications from the same investigation are not independent replications.

Build retained both substantive corrections: alert activation is not actual protective movement, and broad lives-saved language is not interchangeable with the later research's differentiated retrospective outcome categories. The chapter supplies its own stronger-method and NO_USE counter-reading.

The first local assembly copy gained one extra final newline and failed the expected hash check. Build removed that transport-only byte and verified the exact returned hash before repository integration. The author's own earlier link-count checker failure remains in the source custody packet; it did not require a chapter change. Neither correction is hidden as a clean first pass.

Cross-case review is separate from this exact-copy acceptance. The legal chapter had not returned at this checkpoint; no complete three-case acceptance follows from integrating this chapter.

Subsequent cross-review integration

On 26 October, Build incorporated CC's finite review: two defensible planning positions and a route to NIST for reconciling its published summaries. The current chapter is therefore no longer byte-identical to the original above. Original custody and reading limits remain. These additions report no new site assessment, contact or field efficacy.

From refuge research to a local arrangement: a Marin County evidence trail

Follow-on to the Camp Fire case, 26 October 2026. Public-record inquiry only; not a site assessment or live evacuation instruction. The original case and released ME1 remain unchanged.

What the trail actually establishes

Marin provides a real implementation example at the level of published planning, education and institutional ownership. It does **not**, on the evidence inspected here, establish that a particular refuge will protect a particular person, or that NIST caused the local arrangements to be adopted.

The strongest directly readable physical artifact is a one-page Fire Safe Marin refuge leaflet labelled **August 2024**. Its content treats refuge as a contingency and points residents towards local evacuation maps. That is an actual educational output, not an invented community scenario. It does not establish distribution, comprehension or use in an incident. [1]

Fire Safe Marin's current map directory states that partner agencies and municipalities developed and published FireClear maps with local and CAL FIRE grant funding. Its directory contains community-specific links. Separately, the Marin Wildfire Prevention Authority (MWPA) describes a funded cross-jurisdictional detection, alert and evacuation programme and a network of routes and refuge areas. The published financing mechanism is Measure C core funding for the listed programme projects, with additional local funding possible. Neither page is an audited account of spending or maintenance at an individual refuge. [2, 3]

This is more than a recommendation sitting in a research report: named organisations describe work, identify a financing route and expose actual public information. It remains less than a demonstrated end-to-end protective capability.

The causal story had to change

The initial question was whether a community had implemented the NIST guidance. The chronology makes that wording too strong. The checked local leaflet is dated 2024, before the NIST update announced in April 2025. The NIST update must not be credited with creating an already documented local practice. [1, 4]

That does **not** prove Marin developed the practice independently of every NIST publication: earlier NIST Camp Fire studies existed. Nor does chronological precedence settle mutual learning or later revisions. No inspected local decision record explicitly traces the arrangement to the particular NIST recommendation.

The corrected question is: **what local arrangements can be evidenced, how do they relate to the specialist guidance, and which part of the research-to-protection chain remains unproven?** Causal attribution stays UNKNOWN. Similarity is a reason to compare, not a substitute for a provenance record.

Owners and products, with the remaining gap attached

Link in the chain	Public evidence inspected	What can be concluded	What remains unestablished
Specialist knowledge	NIST's April 2025 update explains advance refuge planning and no-notice contingencies	A published specialist planning approach exists	That any named local decision adopted it because of NIST
Local educational output	Fire Safe Marin's August 2024 leaflet	A dated resident-facing refuge explanation exists and is readable	Household receipt, comprehension or ability to act
Mapping and financing	Fire Safe Marin's community map directory names partners and funding sources	A public directory and an attributed mapping programme exist	Grant award identifiers, expenditure, current accuracy or complete geographic coverage
Countywide programme ownership	MWPA's detection/ evacuation programme identifies Measure C core funds and partner roles	A named funding/ coordination institution is visible	A refuge-by-refuge maintenance contract, inspected budget line or current condition record
Technical assessment product	MWPA's ingress/egress assessment page names its lead and technical partners and links reports and a viewer	An existing technical workstream and owner are identifiable	That a model score certifies any refuge or predicts an individual's survival
Residents' official route	Marin County's preparedness page routes readers to alerts, zones and local maps	There is a county-owned public entrance	That every reader can receive a warning or complete the onward journey
Protective effect	No relevant observed incident-outcome evaluation found in this bounded inquiry	No such outcome is claimed	Actual refuge tenability, evacuation success and comparative effectiveness

The MWPA assessment page specifically names MWPA as lead and Sonoma Technology with specialist partners. It identifies communication, transport and route constraints, and explicitly limits the product to planning rather than live incident use. This supplies a concrete handover to an existing domain method, not a reason to build a parallel ME planning tool. [5]

The county's own public preparedness page connects alerts, maps and evacuation-zone information. It is evidence of an information route, not evidence that its recipients have transport or a viable physical route. [6]

Can the public information actually be reached?

One complete leaflet and the Fire Safe Marin map index were readable through this inquiry's tools. Following one listed Inverness map link reached a QR-hosted page exposing template fields rather than a legible map. A listed Novato PDF returned a cache-miss error. A direct Southern Marin Fire evacuation page returned HTTP403, although a search index exposed an account of its refuge arrangements. That indexed account was **not promoted to a freshly verified local inspection**. [2, 7]

Two 2026 CWPP summary links could not be opened by the web tool; the older full plan exceeded its content-length limit. These are retrieval limits, not proof that a human browser cannot open the documents, that the community lacks plans, or that physical arrangements fail. No alternate transport was used to evade a refusal. The tool also could not establish completed map rendering or mobile/accessibility performance.

The 2026 CWPP landing page is itself readable. It reports a 2025 update contract, names modelling partners and describes public engagement, including a May 2026 forum. Those are attributed programme statements, not an independent evaluation of consultation quality or resulting safety. The attached summaries were not read. [8]

The practical distinction is important: **publishing a resource, listing it, retrieving it and being able to act from it are four different accomplishments**. Here, some information was retrieved and some links hit a tool boundary. An actual resident's successful use was not tested. That is a specific ceiling, not a global verdict on the service.

What would close the next link?

The existing actors already have the relevant institutional positions: Fire Safe Marin for its educational/map publications; MWPA for its programme and assessment; county emergency management and local operational authorities for their respective public and incident roles. A site custodian or maintenance owner must be identified in the relevant local record rather than inferred from a programme logo.

For one chosen site, the next useful **documentary** evidence would be its current approved assessment, named upkeep owner, inspection/review date, available capacity and onward-movement arrangements, with evidence of allocated resources. An adopted document is still not a field performance result. Independent exercise or incident evidence could test parts of the arrangement, with explicit scenario and access limitations. Nothing in this inquiry establishes that these records are absent; they were not obtained within this bounded pass.

No new form or repository is required to ask that question. The existing planning owner can supply, qualify or refuse the particular claim. No contact was made, and neither an unsolicited request nor an operational intervention is authorised by this research note.

The useful correction to the original case is therefore constructive and limited: local refuge education, mapping and a funding/assessment structure are already publicly evidenced. ME does not deserve credit for inventing them. The unresolved join is from

these products to a maintained, person-usable arrangement under specified incident conditions. An external practitioner may find the existing specialist account sufficient; **NO_ADDITIONAL_USE_FROM_ME** remains a legitimate result.

Sources and retrieval scope

1. Fire Safe Marin, Temporary Areas of Refuge, one-page leaflet, labelled August 2024. Web tool parsed the complete one-page text. No physical mailing or field use established; no local copy/hash or successful visual rendering claimed.
2. Fire Safe Marin, Evacuation Maps. Directory introduction, funding attribution and listed community routes inspected. The entire county's map contents were not audited.
3. MWPA, Improve Detection & Evacuations. Funding/partner description and listed project types inspected. Programme-level funding is not individual-site expenditure.
4. NIST, Updates Critical Wildfire Evacuation and Sheltering Guidance, 24 April 2025. Comparison source, not an adoption receipt.
5. MWPA, Evacuation Ingress/Egress Risk Assessment. Lead agency, contractors, planning-only boundary and output links inspected; linked report PDFs and GIS model not fully examined.
6. Marin County, Evacuating in a disaster. Short official resident entrance inspected. No live incident status or personal address queried.
7. Failed inspected routes: listed Inverness QR map; listed Novato map; Southern Marin Fire evacuation page. Outcomes are tool-specific observations on 26 October 2026, not general availability findings or emergency advice.
8. MWPA, 2026 Community Wildfire Protection Plan update. Landing-page narrative read; executive and Southern Marin linked summaries failed retrieval. No plan adoption vote or whole-plan review claimed.

Disposition: useful public implementation trail with causal attribution, site-specific maintenance, resident usability and observed protective performance still bounded/UNKNOWN. Optional supporting material for Build's casebook integration; not a replacement case or a new release decision.

Case 3 source and editorial receipt

Framework Build, 26 October 2026. This is a retrieval and editorial record, not an employment field result. Case text: Making a first rung.

Source	Retrieval performed	Use and limit
UK Jobs Plan	Official HTML retrieved 26 October 2026	Dated policy ambition and existing owner route; delivery not established
DSIT/LinkedIn snapshot	Official HTML retrieved 26 October 2026	Observational platform analysis; causal inference restricted
techUK Alliance	Official HTML retrieved 26 October 2026	Existing programme scope, not outcomes or consent
England funding rules v3	PDF text through web retrieval, linked from current official landing page; targeted paragraph searches/ readback	Paragraphs 66.1, 100–103, 107.6, 112.1, 112.11; no complete legal audit, PDF visual/accessibility assessment or current byte hash claimed
CC premise challenge	Complete GitHub comment read	Independent project criticism; not field testimony or a substitute for primary rules
Released ME1.0 source	Exact full CORR2 manuscript had been read during publication; chapter distinctions checked against that source	Frozen conceptual basis; no novelty/ efficacy inference

Review and correction

An internal Build source reviewer, separately identified from CC and Codex Windows, checked the funding scope and causal cautions. It confirmed three necessary boundaries: qualifying training mentorship is conditional; funding rules do not govern every ordinary starter job; policy targets and hiring data do not establish delivery or AI causation. These boundaries appear explicitly in the case. This was an AI source challenge, not peer review.

The initial-reading correction is retained in the chapter: money labelled training plus a feedback form does not establish an affordable supported role. No qualifying mentoring was categorically declared unfundable, and no extra ME reporting duty was introduced.

No participants, businesses, systems, accounts or vacancies were inspected. All employer/task details are constructed. No email, recruitment, registration, pilot, job, payment, intervention or achieved benefit is implied.

Cross-review addition, 26 October 2026

CC's review6426587379 supplied the IfG implementation comparator (whole overview read by CC; separate case publications not read). Build added C6 and a proposed route for recurring funding constraints to the owners of the rules. This is an analogy from pension implementation, not evidence about the constructed employer or a claim that any consultation occurred. CC checked the DfE-to-DWP transfer against the rules' version-3 change log. Build did not repeat that source investigation.

Additional online case studies — source and editorial receipt

27 October 2026 · provenance and evidence limits for Cases 4 and 5. This note is part of the working online casebook, not the dated 26 October 60-page PDF or self-contained HTML snapshot, and not a modification of the frozen Mechanical Ethics v1.0.0 release.

Separate editorial lineage

Case 4 — GitLab, January 2017. The starting source was Codex's completed CASEBOOK-GITLAB-DEEP-01 private technical draft, supplied to Framework Build on 26 October 2026 at `campfire-relay@e62639d940df7efb68594963d13243ac38e75a08/gitlab-deep-case-20261026/GITLAB-RECOVERY-DRAFT.md`: **20,733 UTF-8 bytes, SHA-256** `82fff325be6215f2e2b13b648210743aae90e62ab1db393ccf77efc562e73d0f`, Git blob `186c91ff6e9a941bd386013b4f79d1b785db32c5`; original terminal; Build's specific editorial selection. This edition is **not byte-identical** to the draft. Campfire integrated the existing, material Build corrections: separate **physical candidate feasibility**, **evidence sufficient to rely on a route**, and **authority/risk acceptance**; bring GitLab's reported webhook recovery and identifier sequence advance forward as actual attributed interventions rather than treating only hypothetical side effects. The text adds an explicit comparison against ordinary SRE/incident methods with NO_MATERIAL_DIFFERENCE or EXISTING_METHOD_BETTER possible.

The Codex source receipt describes eight public sources [S1–S8] and exactly which parts were read at that time. On 27 October Campfire separately reopened the two original GitLab blog accounts and the dated public follow-up tracker via public text retrieval; it **did not independently audit full mutable GitLab issue-comment histories, original incident command logs, internal database state, user outcomes or all later remediation tests**. The older tracker descriptions and their checkmarks are not independently dated to 2017. The two GitLab articles are related statements by the same operator, **not independent replications**. Exact return-to-service and lost-work timestamp variations are retained rather than forcibly reconciled.

Case 5 — NLS. Claude Code's 26 October constructive source scout identified a stronger **positive capability** case against US Library of Congress materials and an independent GAO review. The scout was reconnaissance, not a ready chapter. Campfire authored this new chapter on 27 October and directly inspected accessible text of GAO-16-355, Library of Congress NLS overview, network roles, the 2021 Federal Register rule HTML, 2024 NLS programme report and 2025 operator retrospective published in 2026. The case quotes no private patron, extracts no individual health/application record and uses no personal dataset.

The **2016 GAO page** and its follow-up recommendation status are **one federal witness plus later status**, not two independent studies. The **2024/2025 device and programme figures are NLS operator reports**, not field evaluations of their impact or a causal series. The 2021 rule is a documented legal change; no individual enrollment outcome follows simply from its text. The related OIG/annual reports named in CC's scout were not reread in full for this chapter; the chapter intentionally omits annual reader-count trend arithmetic, rather than republish ambiguous cross-year denominators as new independent conclusions.

Case 6? There is no Case 6 in this editorial increment. The file `08-reading-across-five-cases.md` is an **addendum comparing five cases**, not a sixth case.

Exact current outputs and scopes

File	Purpose	Scope
06-gitlab-recovery.md	Historical technical recovery route	Initial/revision and two-copy decision, scoped clocks, reconciliation, specialist comparison
07-nls-reading-access.md	Constructive maintained library capability	Joined public-service mechanisms, audit, changed rules and device distribution, unknown uptake
08-reading-across-five-cases.md	Online cross-case addendum	Restores original three-case distinction while comparing two new domains
Casebook README	Discoverability and edition separation	Original PDF/HTML retain three-case dated snapshot

Source-level and link checks establish this draft's file provenance, not incident replication, causal attribution, legal advice, disability-service usability or empirical advantage from ME. Original three-case texts and original cross-case discussion are unchanged; the online addendum does not rewrite their evidence cutoffs or later representation. The published 60-page PDF, pinned offline HTML, full released v1.0.0 book and rights remain unchanged.

Decision and review boundaries

Editorial integration of these already-public source studies is not an endorsement by GitLab, GAO, NLS, fire-safety experts, accessibility professionals or affected users. The two authorship paths are AI-assisted and source-checkable. The intended tests of practical value remain external: what would existing specialist practice already find, what difference (if any) does ME uncover, and would that difference improve a real, authorised response without moving burden onto someone least able to carry it?

All three results — **EXISTING_METHOD_BETTER**, **NO_MATERIAL_DIFFERENCE** and **MATERIAL_ADDITION** — and **INDETERMINATE** remain permitted. Failure of an ME addition is informative. Do not turn these new case documents into proof that the project has been tested on practitioners.

Constructive-design comparison: first editorial pass

Framework Build, 26 October 2026. This is a bounded comparison of presentation and evidence, not a ranking of casebooks or an effectiveness study.

Primary example inspected

Public Design Evidence Review: Case Study Bank, Cabinet Office / Policy Profession, updated 9 March 2026. Read the introduction and complete HMRC plastic-packaging-tax case in the returned HTML, including its problem, design work and reported outcomes. The other twelve cases and linked documents were not reviewed in this pass.

The introduction identifies a selected collection rather than comprehensive coverage. The HMRC example connects a practical design question to supply-chain work, administrative constraints and subsequent choices. It distinguishes a reported change in demand from a causal claim about the tax. These are useful editorial practices: expose the work between intention and decision, state the selection boundary, and separate an outcome from attribution. The account is an institutional contribution, not an independent evaluation of its own design methods. No figures or substantial source wording are reproduced here.

A London City Hall community-design story appeared in search, but direct opening returned HTTP 403. It was not bypassed or counted as a read comparator. Search snippets were not promoted to a complete case review.

Gap in our first edition

Our chapters provide substantial interpretation before readers have much chance to form their own account. This is Build's editorial diagnosis, not a finding from the comparator's authors. Source precision alone does not establish a good learning experience.

Concrete change

Added optional reader decision exercises that invite an initial judgement, a changed assumption and a counter-reading. They retain the distinction between constructed prompts and historical evidence. They neither supply operational instructions nor claim tested pedagogy. No additional case is added merely to increase the count.

Cross-comparisons with law, engineering incident analysis and public administration are separately assigned to CC and Codex. Those assignments are not completed reviews until their actual returns are read and integrated.

Interim editorial review: returned cases B and C

26 October 2026. A separate internal Build AI reviewer read the complete Camp Fire and paid-first-rung chapters, opening and comparative closing. The reviewer did not author those chapters. This is not external peer review, legal or fire professional review, practitioner validation or complete three-case acceptance. Case A had not returned. No new NIST source audit was performed.

The reviewer found no blocking problem in comparing the two returned cases: the text distinguishes retrospective fire evidence from the constructed employment situation and makes no inference of comparative efficacy.

Build accepted and applied three concrete corrections:

1. Existing institutional programmes are routes to investigate, not established superior alternatives merely because they exist.
2. The closing now says that a training allocation does not establish that all necessary supervisory work is funded or available. This preserves the funding rules' allowance for qualifying training mentorship while rejecting an inference of complete capacity.
3. CC's employment premise challenge is explicitly labelled internal AI work, not left open to interpretation as outside practitioner evidence.

The exact Camp Fire chapter remains unchanged. Its historical outcome and PDF reading limits remain visible. Final review must include the actual legal chapter and the revised whole comparison; this interim review does not replace that task.

Whole-casebook editorial review

26 October 2026. Framework Build assembled the three specialist chapters, a constructive entry example, an opening and a comparative closing. A separate internal Build AI reviewer read all six prose files and the legal source receipt. This was editorial review, not professional legal or fire-safety review, external peer review or an observed test of the framework's usefulness.

Authorship and original custody

- Case 1: Claude Code, returned in COM #877 comments 6425618420 and 6425621989. Build extracted the fenced UTF-8 text and verified both original hashes: chapter `452de92b6ba5077cd8b719b03b37b5bdb5a599d8ff468a551bd814c4ac1c8955` (33,575 bytes); receipt `399ebf73d7d2e8acb7b81ef3dc9e4073130502b57dbc73f50e07a43f493efb83` (8,415 bytes). The integrated chapter has the editorial changes below and is not claimed byte-identical. Original returns remain available at those links.
- Case 2: Codex Windows. Its original chapter and later editorial additions are distinguished; see the integration record.
- Case 3, entry page, opening and closing: Framework Build. The earlier B/C review records corrections already applied.

Whole-case findings and changes

The reviewer found the constructed entry clearly distinguished from observation and the B/C comparison appropriately bounded. The remaining assembly-status text was stale; Build replaced it with links to all three actual chapters and a comparison that includes the legal case's distinct contribution.

Build and the reviewer identified these legal-case corrections:

1. A deliberately misleading initial telling is no longer described as containing truth in every sentence.
2. The earlier IRS manual's individual handling is identified as administrative practice, not a separate statutory requirement.
3. Facial compliance of the agreement does not establish compliance of requests or the implementing procedure. The synthesis no longer locates the whole defect in software or equates performed checks with substantive compliance.
4. TIGTA's lack of independent validation of DHS data describes review scope, not demonstrated inability to see that data.
5. Restricting future disclosure is distinguished from proving protection; the November court restriction is not placed after the February admission.

6. Current-sounding litigation and remedy statements are tied to the chapter's source cutoff, 17 September 2026. No later docket status was established.
7. Numerical comparison is omitted for lack of an evidenced common scale and counterfactual, rather than because a particular symbol is absent from ME1.

Build additionally inspected 26 U.S.C. §7431(a)–(e) on 26 October. The chapter now recognises that damages depend on liability conditions and exceptions, and that subsection (e) contains specified mandatory notification triggers. Whether those triggers applied here remains unknown. This corrects an overly discretionary notification proposal without deciding the legal outcome. Build also read the official TIGTA report's opening findings; no independent full legal-source audit or fresh docket investigation is claimed.

The closing now distinguishes the three evidential settings: preliminary legal findings within a dated record, retrospective fire research, and a constructed employment situation against documented policy. Shared reasoning questions do not imply shared legal, physical or employment thresholds, nor prove efficacy.

Remaining limits

The legal chapter's partial source reads and image-only pages remain disclosed. The fire chapter's full-PDF access limits remain disclosed. Neither constructed example reports actual participants or outcomes. No lawyer, emergency planner, employer or entrant has validated the assembled supplement. These limits bound its claims; they do not prevent readers from using and challenging the reasoning.

The separately commissioned refuge implementation trail is optional follow-on work, not a dependency of this casebook. No adoption or implementation benefit is inferred merely from similarity to NIST guidance.

The supplement adds only casebook files. It does not revise the frozen v1.0.0 book, figures, tag, release or rights. Repository readback and local link checks establish delivery integrity, not factual truth or field effectiveness.

Second editorial wave: owned repair, 26 October 2026

Build adopted the principal engineering correction from Codex's cross-case review. Section 12 now distinguishes legal sufficiency, implemented checks and authority to disclose. Its proposed work names responsible roles, evidence of completion, confidentiality limits and what a successful check would still not establish. This replaces the vague instruction to rebuild and publish; it does not report an actual repair or authorise disclosure.

The complete private handoff was read at immutable commit `ea7db74ef9d764a1cfc4c363e76d113ba8a2b96d`, file `casebook-v02-cross-case-01/CROSS-CASE-HANDOFF.md`, SHA-256 `e9a9d6165be66df0fe2f30e4e4a49bbb62446aa4cbe66b7b7a2155c6ee01dee4`.

Codex's comparator was the Google SRE Workbook's actual, anonymised satellite-diskerase postmortem. This records Codex's primary-source review, not a separate Build reading of that comparator or a legal authority.

Two further wording questions remain with CC within its existing finite review: distinguishing preprocessing from branch matching and substantive sufficiency; and qualifying the proposed blanket closure of safeguard findings. They are not yet integrated here. CC's separate review of the fire and employment cases also remains pending. This is an intermediate improvement, not completion of the cross-case editorial wave. Original source receipts and cutoff remain.

Cross-review closed into this edition

Both finite specialist returns are now integrated: Codex 6426277846 and CC 6426587379. The pending statements above describe the earlier checkpoint, not current review status.

Build accepted all three bounded findings from each reviewer. Case A separates format screening, branch matching and substantive sufficiency, including unnecessary rejection; CC confirmed the distinctions from its saved sources. The blanket safeguard proposal is replaced with assessment under applicable duties and the statutory decision-maker. Court reasoning is attributed and paraphrased; the September cutoff is unchanged. Case B now asks the reader to weigh pre-designation against misplaced reliance and names NIST as the owner of its differing summaries. Case C routes recurring funding constraints to rule owners and includes the implementation lesson about hearing from actual administrators. No contact occurred.

CC read the FJC flag-salute teacher handout whole and selected sections of its larger packet, not a law-school casebook. Its IfG comparator was the whole overview report, not the four separate case publications or a classroom teaching note. Failed retrievals remain in its public receipt. These are useful partial substitutes, not fulfilment of stronger source claims. Build read the entire specialist returns, not an independent repeat of their comparator research. The substantive editorial wave is complete; external reader validation, final merge approval and reader browser QA remain separate.

Technical revision requested by Mark

Mark rejected the guided-reader style and requested a dry, substantially detailed scientific/engineering supplement. Build replaced the entry and comparative closing, removed staged-choice instructions, revised the chapter framing and added three analytical sections: disclosure-path coverage and error denominators; protective-journey timing and causal identification; and supervision capacity, cost allocation and learning measurement. These are explicitly authored models, not new observations or fitted predictions. No empirical values were invented.

Earlier exercises and the introductory scene remain in source history and as superseded files, but are excluded from the technical reading sequence. Existing specialist evidence and its limits remain. The employment chapter is explicitly a constructed design study,

not a third historical case. This is a first technical revision; it does not close the documented gaps in full-report reading, field data or professional assessment. Original cross-review receipts still support the underlying corrections, not acceptance of this later presentation.

26 October — Campfire technical correction and integrated reading order

Supersedes the model/structure claims in the first technical revision at 1013a7a74cef94596a16668bc360ec99d3809970. Campfire's review, COM #877 c6427214751, identified a material error in the fire timing condition.

- Fire: replaced the general arrival-before-any-closure condition with traversal intervals for each route segment, including waiting, and continuous destination occupancy requirements. The old scalar test is identified only as a conservative special-case screen. No numerical thresholds, field validation or new research.
- Employment: retained the aggregate hours constraint but added competent capacity for each time slot, uninterrupted/response-time needs, incompatible assignments, contingent demand and fallback. Funding, capacity and learning remain separate.
- Disclosure: r now denotes a submitted row with its request-level context, not a person or output record. Branch-specific evidence and unknown coverage are mapped to substantive conditions; unresolved legal classifications remain unresolved.
- Structure: models now follow their relevant mechanisms and evidence. Employment cost and learning analysis is placed with funding and observation respectively. Competing interpretations precede correction decisions. Repetitive introductory framing was reduced without replacing the original source receipts or cutoff.

This is one Build-owned correction pass, not a new specialist review wave. Campfire retains the merge decision. The original failed fire formulation remains in Git history and this correction record. Browser/mobile/keyboard verification remains NOT_RUN; source and deterministic reader checks are reported separately.